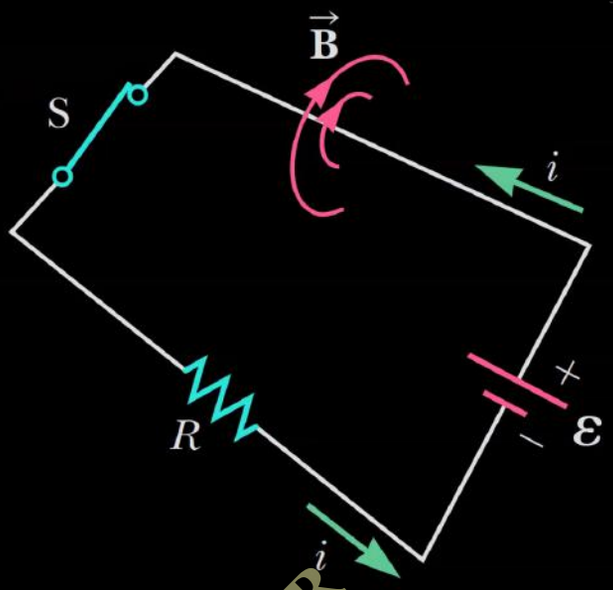
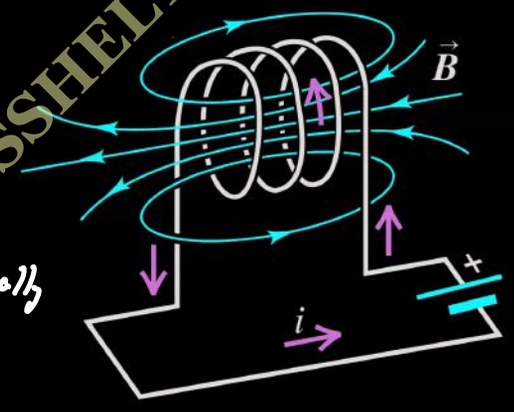
Self Induction:

Whenever a loop carries a current, it produces a magnetic field through itself (see figure), if this field changes with due to changing current, it produces an EMF in the loop (due to its own changing current). This phenomenon of generation of EMF in a loop due to its own current is called self induction.



Self Inductance (or just Inductance L):

The induced EMF in a coil is proportional to rate of change of current through it, the constant of proportionality is called Inductance of the coil (or self inductance). Mathematically $\mathcal{E} = -L\frac{di}{dt}$, the negative sign appears in accordance with Lenz's law.



If the current in the loop is increasing, the EMF induced opposes it. Now by Faraday's law: $\mathcal{E} = -d\phi/dt$ thus by comparison, $L\,di = d\phi$, thus integrating $Li = \phi$ + constant, however when current is zero, there is no flux thus, $\phi = Li$ or self inductance of any coil can be alternately defined as flux per unit current through the coil i.e.

$$L = \phi/i$$

For a multiturn coil if ϕ is flux through one turn and N is the number of turns then $L = N\phi/i$

Self Inductance Of A Long Solenoid: Consider a solenoid of radius R, length l and N number of turns then, $B = \mu_0 \frac{N}{l} i$ thus flux/turn: $\mu_0 \frac{N}{l} i \times \pi R^2$, thus total flux linkes:

$N\phi = \mu_0 N^2 i \pi R^2/l$ thus $L = N\phi/i$:

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$L = \mu_0 N^2 \pi R^2 / \ell$ or $\mu_0 N^2 A / \ell$.

$$\boxed{L = \mu_0 N^2 A / \ell}$$

→ Here we assumed an air core solenoid. If instead of air the solenoid were filled with a material of permeability μ then,

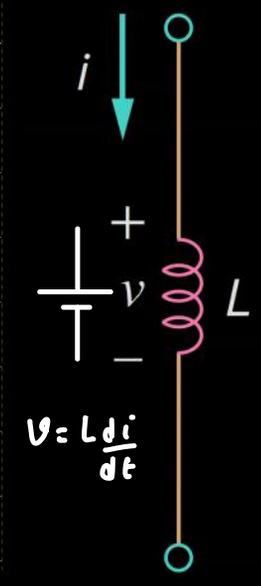
$$\boxed{L = \mu N^2 A / \ell}$$

→ Sometimes inductors are also known as coils or chokes.

→ SI unit of inductance is Henry (H), often we will come across inductors in milli Henries (mH).

Inductor As A Circuit Element:

The adjacent figure shows circuit symbol for an inductor. Since an inductor produces a current dependent EMF $\mathcal{E} = L\,di/dt$, for purpose of applying loop law to the circuits, an Inducton is replaced by a battery of EMF $\mathcal{E} = L\,di/dt$ against the current in the branch. It does not matters whether the current is increasing or decreasing. For instance in the shown figure if $L = 1H$ and $\frac{di}{dt} = -1\ A/sec \Rightarrow \frac{L\,di}{dt} = -1\ Volt$,



now a battery of -1 Volt against the current means the same thing as a battery of $+1$ Volt along the current.

Pure Inductor In A Circuit:

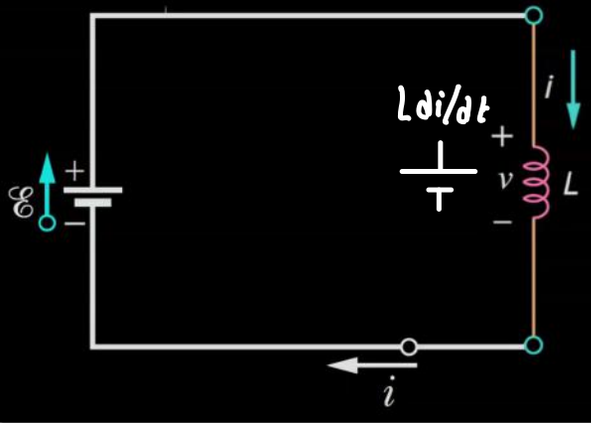
Applying loop law to the adjacent circuit:

$$-\mathcal{E} + L\,di/dt = 0$$

$$\Rightarrow L\,di = \mathcal{E}\,dt \Rightarrow L\int_0^i di = \mathcal{E}\int_0^t dt$$

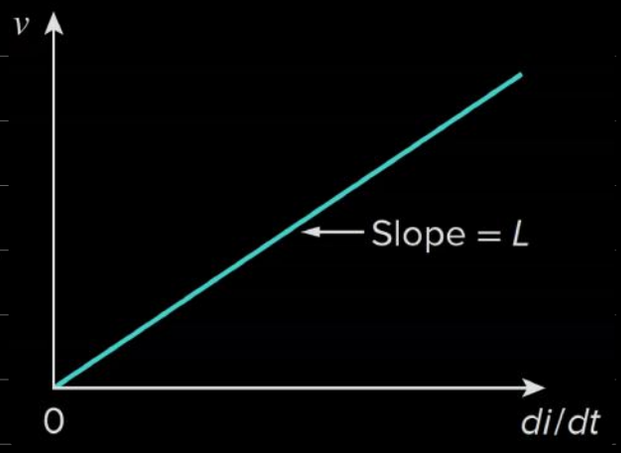
$$\Rightarrow Li = \mathcal{E}t$$

$$\Rightarrow i = \frac{\mathcal{E}}{L}t$$



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Thus in a pure inductor circuit, the current will keep on increasing for even linearly. The work done by the battery is stored in the form of magnetic Energy in the inductor. (Unlike resistor, an inductor does not dissipate any heat).

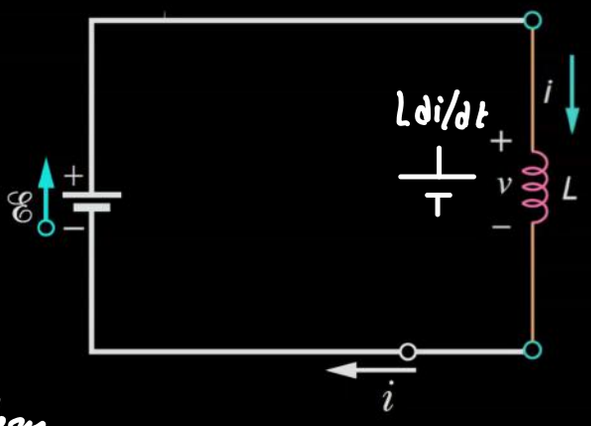


→ Since $\mathcal{E} = L\,di/dt$ an inductor does not allow sudden changes in current under normal circumstances because if i changes abruptly, there will be infinite EMF in the inductor. (This is similar to saying that velocity of a mass cannot change abruptly because it will require infinite force, although with impulsive forces we do see very large accelerations).

Energy Stored In An Inductor:

Let us once again write the loop law equation for the inductor circuit:

$$\mathcal{E} - L\frac{di}{dt} = 0 \Rightarrow \mathcal{E} = L\,di/dt$$



Now multiplying both sides by i:

$\mathcal{E}i = Li\,di/dt$, now recognize that LHS is the power delivered by the battery therefore RHS must be the rate of storage of Energy in the inductor thus:

$$\frac{dU}{dt} = Li\frac{di}{dt} \Rightarrow dU = Li\,di \Rightarrow \int_0^U dU = \int_0^i Li\,di$$

$$\Rightarrow \boxed{U = \frac{1}{2}Li^2} \leftarrow \text{Energy Stored in an inductor.}$$

Energy Density In Magnetic Field: Recall that we calculated inductance of a long solenoid as $L = \mu_0 N^2 A/\ell$, now to find the energy density associated with magnetic field we will divide the total energy stored by the volume of the solenoid. Thus

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Energy density: $u = U/Al = \frac{1}{2} \times \frac{\mu_0 N^2 A}{l} \times i^2 \times \frac{1}{Al}$

$\Rightarrow \quad u = \frac{1}{2}\mu_0 n^2 i^2 = \frac{(\mu_0 ni)^2}{2\mu_0} = \frac{B^2}{2\mu_0}$

Thus energy density associated with magnetic field:

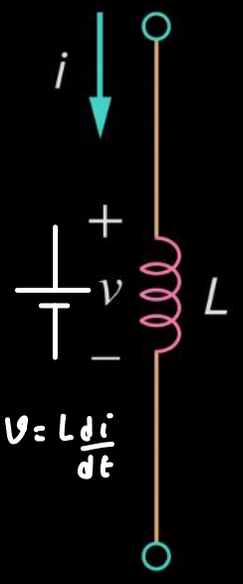
$$\boxed{u = B^2/2\mu_0}$$

→ Even though we proved this result using the special case of a solenoid, the same expression is valid everywhere in free space. (Just as we proved for electric fields $u = \frac{1}{2}\epsilon_0 E^2$ using capacitor).

Current Through An Inductor By Integrating Voltage:

Note that an ideal inductor has zero resistance. Also recall that $\rho \vec{J} = (\vec{E} + \vec{E}^*)$, now since $\rho = 0$ we can say that $\vec{E} + \vec{E}^* = 0$ thus $\vec{E}^* = -\vec{E}$ and hence,

$$\int \vec{E}^* \cdot d\vec{l} = -\int \vec{E} \cdot d\vec{l}$$ thus EMF and



potential difference across an ideal inductor are exactly equal, net electric field in an ideal inductor is always zero (However real inductors will also have some coil resistance). For the shown inductor, non conservative field tries to push the charges against the current ∴ conservative field pushes the charges along the current thus for an increasing current, lower side is at lower potential. (Along electrostatic field lines potential decreases).

Thus:

$$V = L\frac{di}{dt} \Rightarrow \int_{i(t_0)}^{i} di = \frac{1}{L}\int_{t_0}^{t} V\,dt$$

Thus $$\boxed{i = i(t_0) + \frac{1}{L}\int_{t_0}^{t} V\,dt}$$

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EXAMPLE: Find the current through a 5H inductor if, for $t<0$ $v(t)=0$ and for $t>0$, $v(t)=30t^2$. Also find the energy stored at $t=5$ sec and power delivered to the inductor as a function of time.

Solution:

$$i = i(t_0) + \frac{1}{L}\int_{t_0}^{t} v(t)\,dt = 0 + \frac{1}{5}\int_{0}^{t} 30t^2\,dt = 2t^3 \text{ Amp.}$$

Now $$U(5) = \frac{1}{2}Li^2 = \frac{1}{2}\times 5\times(2\times 5^3)^2 = 156.25 \text{ kJ}$$

Power $$P = vi = 30t^2 \times 2t^3 = 60t^5 \text{ Watt}$$

INDUCTORS IN SERIES:

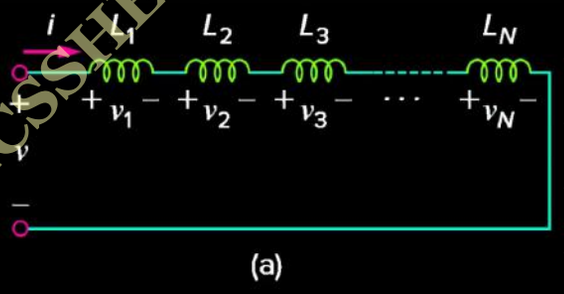
A system of inductors is said to be equivalent to a single inductor if for same rate of change of current they have same EMF. Now consider the adjacent figure:

$v = v_1 + v_2 + v_3 + \cdots v_N$ thus:

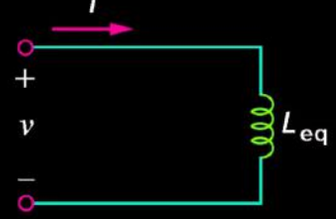
$$v = L_1\frac{di}{dt} + L_2\frac{di}{dt} + \cdots L_N\frac{di}{dt} = L_{eq}\frac{di}{dt}$$

Thus:

$$\boxed{L_{eq} = L_1 + L_2 + L_3 + \cdots L_N}$$



(a)



(b)

INDUCTORS IN PARALLEL:

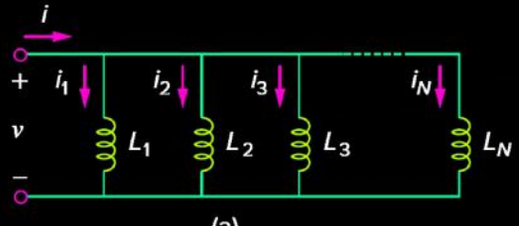
Consider the adjacent figure:

$i = i_1 + i_2 + i_3 + \cdots i_N$

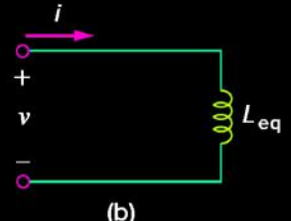
Also $$v = L_1\dot{i}_1 = L_2\dot{i}_2 = \cdots L_N\dot{i}_N$$

$$\Rightarrow \dot{i} = \dot{i}_1 + \dot{i}_2 + \cdots \dot{i}_N = \frac{v}{L_1} + \frac{v}{L_2} + \cdots \frac{v_N}{L_N}$$

Now $$v = L_{eq}\dot{i}$$



(a)



(b)

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$$\Rightarrow \quad v = L_{eq}\left(\frac{v}{L_1} + \frac{v}{L_2} + \frac{v}{L_3} + \cdots \frac{v_n}{L_n}\right)$$

Thus: $$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots \frac{1}{L_n}}$$

→ In parallel, changes in currents are in the inverse ratio of inductances, this can be easily shown: $L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt} \Rightarrow$

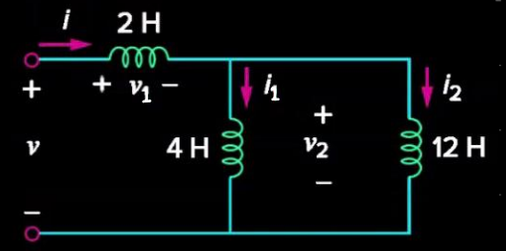
$di_1/di_2 = L_2/L_1$, now since these ratios hold for every incremental current, it also holds for total changes in currents thus: $\Delta i_1/\Delta i_2 = L_2/L_1$.

→ In series potential drops are in direct ratios of inductances.

→ For circuit loops containing pure inductors, loop law will simply become $\Sigma L_i \Delta i_i = 0$ (Currents along the loop are taken as positive).

EXAMPLE:

For the circuit in Figure , $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.



Solution: (a) $i_1 + i_2 = i \Rightarrow i_1 = i - i_2$ @ $t=0$, $i = 4$mA, $i_2 = -1$mA

$\Rightarrow i_1 = 4 - (-1) = 5$mA Ans.

(b) $v(t) = L_{eff}(di/dt)$, Now $L_{eff} = [12 \times 4/(12+4)] + 2 = 5$H

$i = 40e^{-10t}$ mA/sec thus $v(t) = L_{eff} \times i = 200 \times 10^{-3} e^{-10t}$ Volt

thus $v(t) = 200e^{-10t}$ mV Ans

Now $v_1(t) = 2H \times i = 80 \times e^{-10t}$ mV, $v_2 = v - v_1 = 120 \times e^{-10t}$ mV

(c) Now changes in i_1 and i_2 at a general time can be found by dividing the change in i in inverse ratio of inductances and then we can superpose the initial currents to get the net current. Now $\Delta i = [4(2 - e^{-10t}) - 4]$ mA (∵ @ $t=0$, $i = 4$mA)

$\Rightarrow \quad \Delta i = 4(1 - e^{-10t})$ mA

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Now $\Delta i_1 = \frac{3}{4} \times \Delta i = 3(1 - e^{-10t})$ mA

$\Delta i_2 = \frac{1}{4} \times \Delta i = (1 - e^{-10t})$ mA

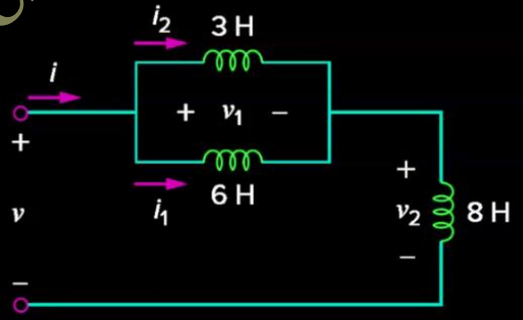
Now $i_1 = i_1(0) + \Delta i_1 = 5 + 3(1 - e^{-10t})$ mA $= 8 - 3e^{-10t}$ mA Ans.

$i_2 = i_2(0) + \Delta i_2 = -1 + (1 - e^{-10t})$ mA $= -e^{-10t}$ mA Ans.

→ For part (c) notice the solution very carefully. Unlike resistors, here currents are not simply divided in the inverse ratio of inductances rather change in currents are in inverse ratio of inductances and we need to carefully work out and add initial currents to obtain final currents.

EXAMPLE:

In the circuit of Figure , $i_1(t) = 600e^{-2t}$ mA. If $i(0) = 1.4$ A, find:
(a) $i_2(0)$; (b) $i_2(t)$ and $i(t)$; (c) $v_1(t)$, $v_2(t)$, and $v(t)$.



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Solution: (a) $i_1(0) = 600$ mA $\Rightarrow \Delta i_1 = 600(e^{-2t} - 1)$ mA

$\Rightarrow \Delta i_2 = 1200(e^{-2t} - 1)$, Now @ $t = 0$, $i = 1.4$ A, $i_1 = 600$ mA

$\Rightarrow i_2(0) = 1400 - 600 = 800$ mA Ans.

(b) Now $i_2(t) = i_2(0) + \Delta i_2 = 800$ mA $+ 1200(e^{-2t} - 1)$ mA

$\Rightarrow i_2 = -0.4 + 1.2e^{-2t}$ A Ans.

(c) $v_1(t) = 6\text{ H} \times i_1' = 6 \times (-1200 \times e^{-2t} \times 10^{-3}) = -7.2e^{-2t}$ Volt Ans.

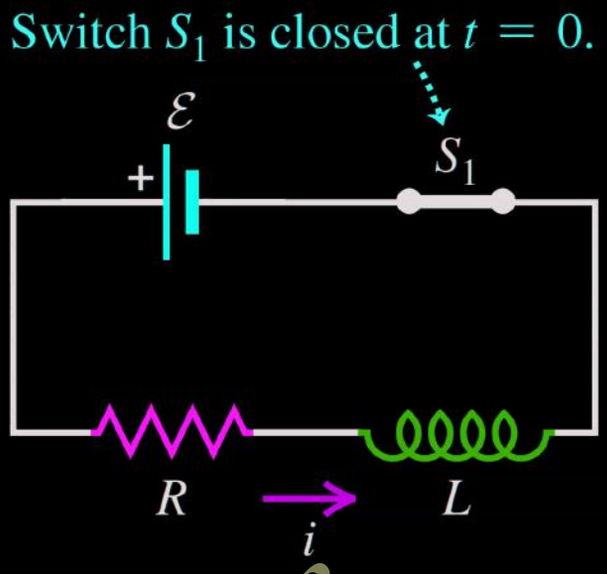
$v_2(t) = 4v_1$ (∵ L of 3H || 6H = 2H & 8H = 2H × 4).

$\Rightarrow v_2(t) = -28.8e^{-2t}$ Volt. Ans.

Now $v = v_1 + v_2 = 5v_1 = -36e^{-2t}$ Volt Ans.

LR Growth Circuit:

Consider the RL circuit shown in the adjacent figure. Let current at $t=0$ be $i=0$ and now the switch is closed. Now applying loop law:



$$iR + L\,di/dt - \mathcal{E} = 0$$

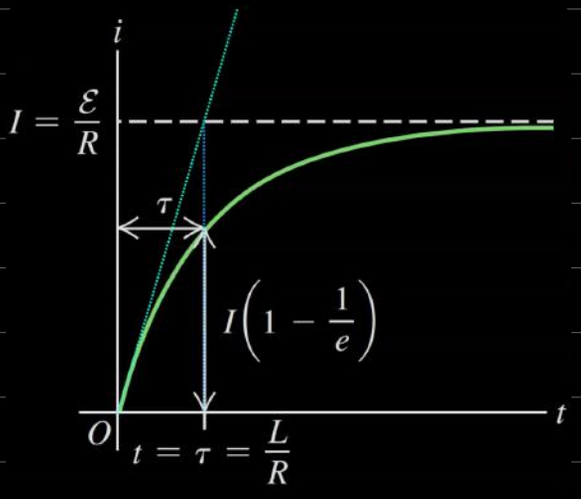
$$\Rightarrow \int_0^i \frac{L\,di}{\mathcal{E} - iR} = \int_0^t dt$$

$$\Rightarrow \frac{-L}{R}\left[\ln(\mathcal{E}-iR)\right]_0^i = t \Rightarrow \ln\left(\frac{\mathcal{E}-iR}{\mathcal{E}}\right) = -tR/L$$

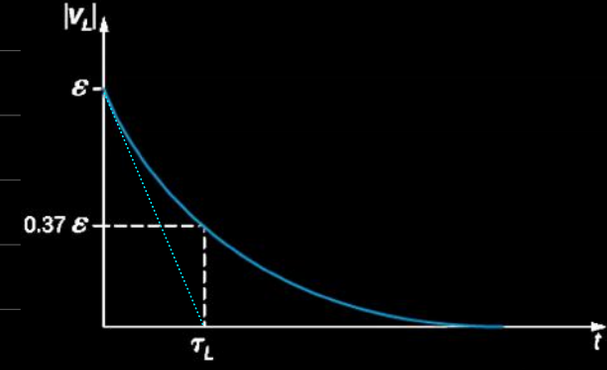
$$\Rightarrow \mathcal{E} - iR = \mathcal{E}e^{-tR/L} \Rightarrow i = \frac{\mathcal{E}}{R}\left(1 - e^{-tR/L}\right)$$

Thus: $\boxed{i = i_\infty\left(1 - e^{-t/\tau}\right)}$ where $\boxed{\tau = L/R}$ and $\boxed{i_\infty = \mathcal{E}/R}$

→ Adjacent figure shows growth of current vs time graph. It can be seen that initial $\frac{di}{dt} = \frac{\mathcal{E}}{L}$, if the current were to constantly grow at this rate, in a time τ, the current would be $\frac{\mathcal{E}}{L} \times \frac{L}{R} = \mathcal{E}/R$, thus physical significance of time constant is that it is the time in which inductor would achieve it full current if the current were to grow at its initial rate.



→ Graph of inductor voltage vs current is shown in adjacent figure. The tangent at initial point intersects at τ. Thus τ is the time when voltage across inductor would go to zero, had it continued to decline at the initial rate.



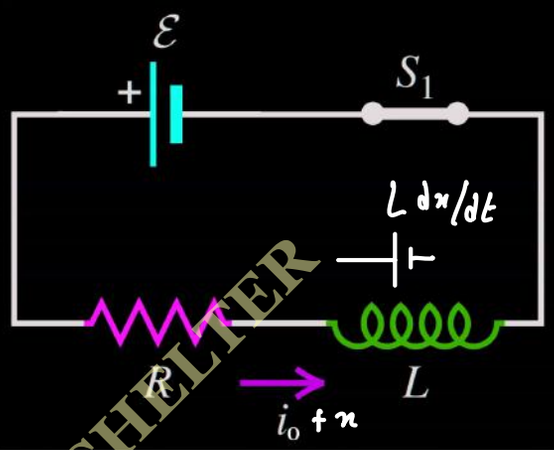
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→ At $t=0$, $i=0$ and @ $t=\infty$ $i=\mathcal{E}/R$ thus at $t=0$ inductor behaves as an open circuit, while in steady state it behaves like a short circuit.

Inductor Growth Circuit With Non Zero Initial Current:

Consider an LR circuit with initial current being non-zero (say equal to i_0), let x be the additional current flown by a time t. Then differential equation governing additional current is:

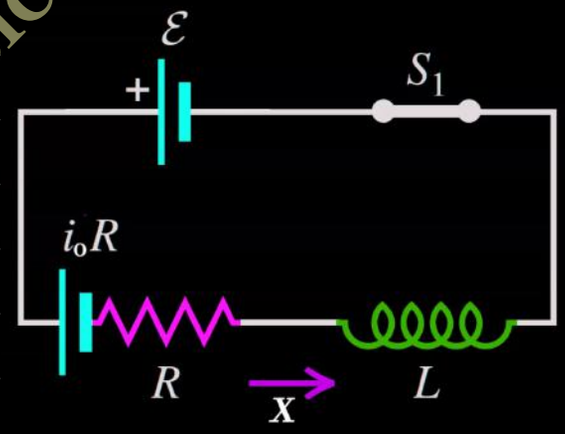
$$(i_0+x)R + L\frac{dx}{dt} - \mathcal{E} = 0$$



Now consider an alternate circuit where the resistor has an added battery of i_0R and the net current in the circuit is x. Then differential equation governing the circuit is:

$$i_0R + xR + L\frac{dx}{dt} - \mathcal{E} = 0$$

$$\Rightarrow (i_0+x)R + L\frac{dx}{dt} - \mathcal{E} = 0$$



This gives us exactly the same differential equation as our actual circuit for purpose of calculation of additional current. Thus for purpose of calculation of additional current a resistor can always be broken in to a battery and a resistor with zero initial current. Once the additional current is figured out, it can be superposed on the initial current to get the net current in the circuit.

→ Rate of heat generation should be calculated carefully, it is not x^2R rather it is $(i_0+x)^2R$.

→ Current at a general time can be thought of as: $i = i_\infty + (i_0 - i_\infty)e^{-t/\tau}$

→ Similarly Voltage across any branch can be thought of : $V = V_\infty + (V_0 - V_\infty)e^{-t/\tau}$

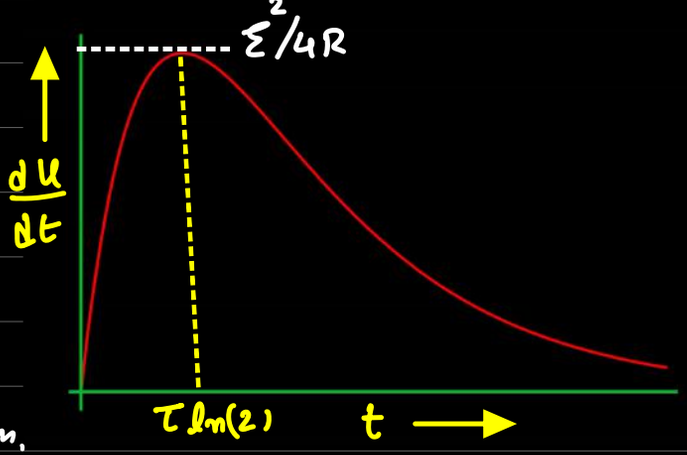
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Maximum Energy Storage Rate In Growth Circuit:

Recall that $U = \frac{1}{2} L i^2$

$\Rightarrow \frac{dU}{dt} = L i \frac{di}{dt}$, Now:

$i = i_\infty (1 - e^{-t/\tau})$ and $\frac{di}{dt} = \frac{i_\infty}{\tau} e^{-t/\tau}$



Thus max energy storage rate occurs when $e^{-t/\tau} \times (1 - e^{-t/\tau})$ is maximum.

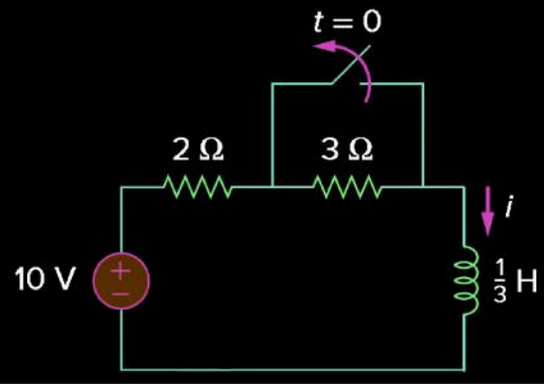
Now $x \times (1-x)$ is maximum when $x = 1/2$ (AM > GM)

Thus for max $\frac{dU}{dt}$, $e^{-t/\tau} = 1/2 \Rightarrow t = \tau \ln(2)$.

At this moment, $\frac{dU}{dt} = L i_\infty^2 / \tau = \left(\frac{\mathcal{E}}{2R}\right)^2 \times L \times \frac{R}{L} = \mathcal{E}^2/4R$.

Example:

In the shown circuit, the switch was closed since a long time and then at $t=0$, it was opened. Find current through the inductor as a function of time.



Solution: Initial current at $t=0$, $i(0) = 10/2$

$\Rightarrow i(0) = 5$ A. Now $i_\infty = 10/5 = 2$ A.

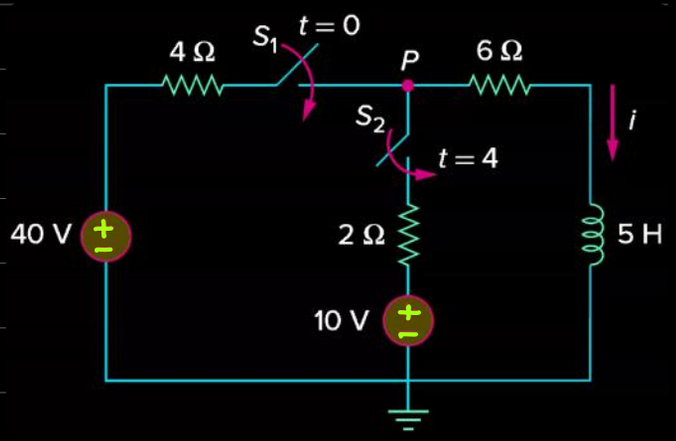
Time Constant $\tau = L/R = \frac{1}{3}/5 = 1/15$ sec.

Now $i = i_\infty + (i_0 - i_\infty) e^{-t/\tau} = 2 + (5-2) e^{-15t} = 2 + 3e^{-15t}$ Ans.

→ So far we have dealt with simple series L-R circuits, if however we have a circuit containing several resistors and batteries we can always thevenize the inductor to simplify the circuit and work backwards to find currents and voltages in the original circuit.

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EXAMPLE: In the circuit shown in the figure, switch S_1 is closed at $t=0$ and S_2 is closed at $t=4$ sec. Find $i(t)$ for all times.



Solution: For $t<4$ sec, $R=10$, $L=5$

$\Rightarrow \tau = 1/2$, $i_\infty = 40/10 = 4A$

thus for $t<4$:

$$i = 4 + (0-4)e^{-2t} = 4(1-e^{-2t})$$ Ans.

Now $i(4) = 4(1-e^{-8})$, Now $\mathcal{E}_{th} = \dfrac{10/2 + 40/4}{1/2+1/4} = 20$ V, $R_{th} = \dfrac{4}{3} + 6 = \dfrac{22}{3}\ \Omega$

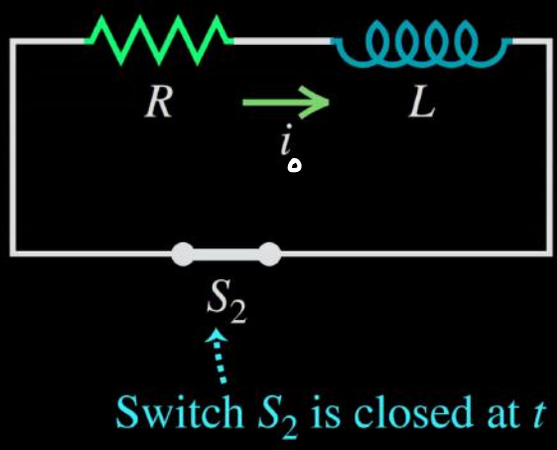
Thus $i_\infty = 20/(22/3) = 30/11$ Amp. Also we are not starting our clock at $t=4$, so we need to take that in to account. Let us assume starting the clock at $t=4$ and later we will appropriately time shift the obtained function. Thus

$$i = \frac{30}{11} + \left(-\frac{30}{11} + 4(1-e^{-8})\right)e^{-22t/15}, \quad e^{-8} \approx 0$$

Now time shifting, $i \approx \dfrac{30}{11} + \left(-\dfrac{30}{11} + 4\right)e^{-22(t-4)/15}$

RL DECAY CIRCUIT:

Consider the circuit shown in the figure.



Switch S_2 is closed at $t = 0$.

For the shown circuit,

$$iR + L\frac{di}{dt} = 0$$

$$\Rightarrow \frac{di}{i} = \frac{-R}{L}dt \Rightarrow \int_{i_0}^{0}\frac{di}{i} = -\frac{R}{L}\int_0^t dt$$

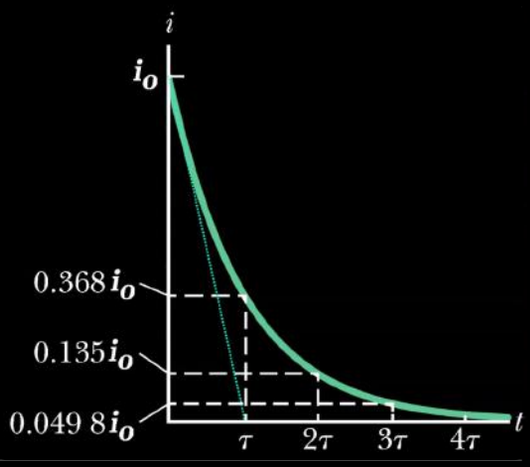
$$\Rightarrow i = i_0 e^{-tR/L}$$

We could have arrived at this result directly also using $i = i_\infty + (i_0 - i_\infty)e^{-tR/L} = 0 + (i_0 - 0)e^{-t/\tau}$

Thus $\boxed{i = i_0 e^{-t/\tau}}$ where $\boxed{\tau = L/R}$

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→ Current vs time graph for the LR decay circuit is shown in the figure. The current exponentially decays to zero. That tangent at initial point cuts the time axis at τ. This means that if current decayed at the initial rate constantly, it would have gone to zero in a time τ.



Important Characteristics Of Basic Circuit Elements:

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v (in series RC)	i (in series RL)

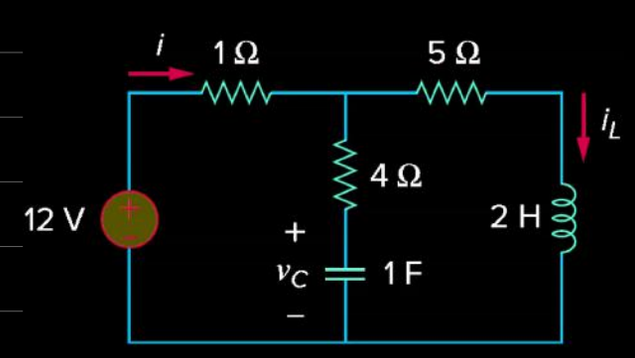
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→ In steady state, capacitor behaves just like an open circuit and inductor behaves short circuit. (Steady state is also called DC Response).

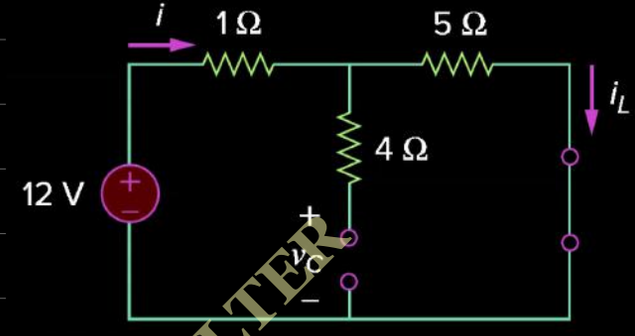
→ Immediately after starting a circuit, capacitor behaves like short circuit and an inductor behaves like an open circuit.

Example:

Consider the circuit shown in the figure. Under DC conditions find: (a) i, V_c and i_L (b) Energy stored in capacitor and inductor.



Solution: In steady state we replace capacitor by open circuit and inductor by short circuit and obtain a purely resistive circuit as shown in the adjacent figure.



Now (a) $i = 12/6 = 2A$ Ans.

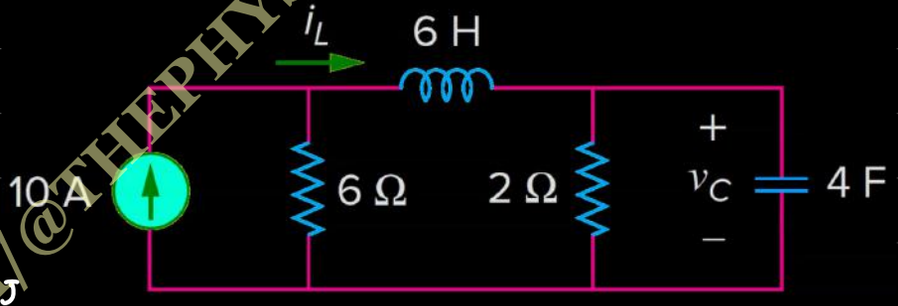
Now $V_c = i \times 5\Omega = 10$ Volt Ans.

$i_L = i = 2A$ Ans.

(b) $U_c = \frac{1}{2} C v^2 = \frac{1}{2} \times 1 \times 10^2 = 50$ J Ans

$U_L = \frac{1}{2} L i_L^2 = \frac{1}{2} \times 2 \times 2^2 = 4$ J Ans

Think: Determine V_c, i_L and the energy stored in capacitor and inductor under DC conditions.



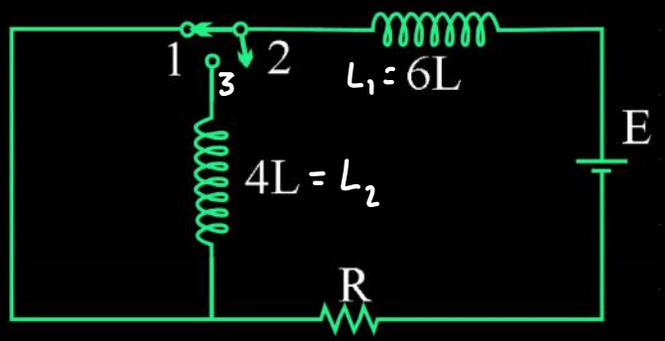
Ans: $V_c = 15$ V, $i_L = 7.5$A, $U_c = 450$ J

$U_L = 168.75$ Joules.

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Forced Abrupt Current Changes In Inductor Circuits:

Consider the circuit shown in adjacent figure. Suppose the circuit was in a steady state. Now what happen if point 2 were connected with 3 and disconnected from 1. Prior to switching current in 6L inductor was E/R while in 4L inductor it was zero, but now suddenly both of them come in series. Neither of them would like to change their current, however due to constraint of being connected in series both of them have to carry the same current. To satisfy the loop law in such case, both inductors must develop an infinite EMF



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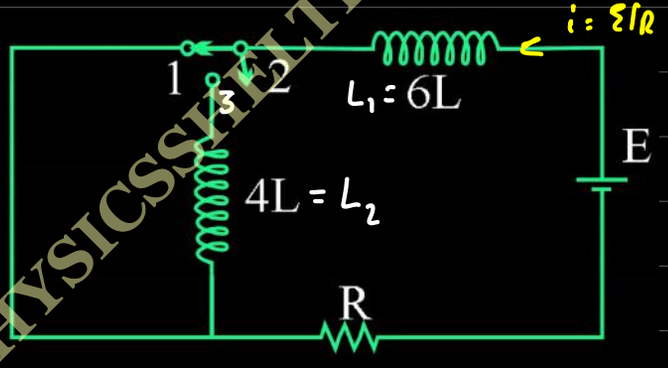
such that the infinities cancel each other. (This is somewhat like inelastic collision of blocks where blocks have to suddenly achieve infinite accelerations, but their internal impulses cancel out). In that case, suppose switching process takes a Δt time and Δi_1 and Δi_2 are currents in the inductors L_1 and L_2 then $L_1\frac{\Delta i_1}{\Delta t} + L_2\frac{\Delta i_2}{\Delta t} = 0$

$\Rightarrow L_1\Delta i_1 + L_2\Delta i_2 = 0$, In general if there were several inductors then,

$$\Sigma L_i \cdot \Delta i_i = 0.$$

EXAMPLE:

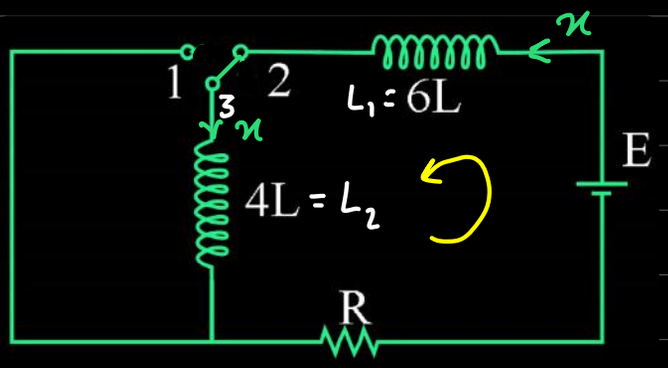
The shown circuit is in steady state. Now 2 is connected with 3 and disconnected from 1 immediately afterwards. Find current in the inductors immediately after this switching process.



Solution: Applying loop law along the shown loop, $\Sigma L_i \cdot \Delta i_i = 0$

$\Rightarrow 6L\left(x - \frac{\mathcal{E}}{R}\right) + 4L(x - 0) = 0$

$\Rightarrow 10x = 6\mathcal{E}/R \Rightarrow x = 3\mathcal{E}/5R$ Ans.

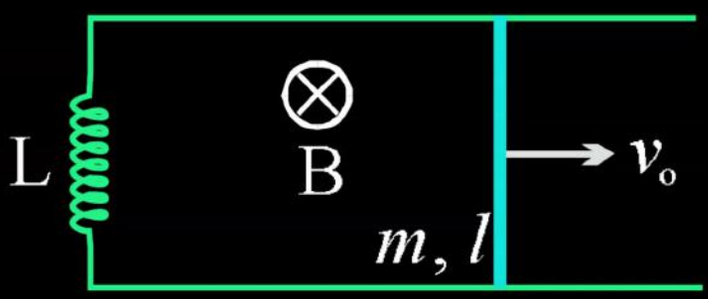


→ This is just like an inelastic collision with analogues: $L_1 \to m_1$ $L_2 \to m_2$ $i_1 \to v_1$ $i_2 \to v_2$ $\phi \to mv$

(In inelastic collision $\Sigma m_i \Delta v_i = 0$, so essentially maths is just parallel).

INDUCTOR SLIDER ARRANGEMENT:

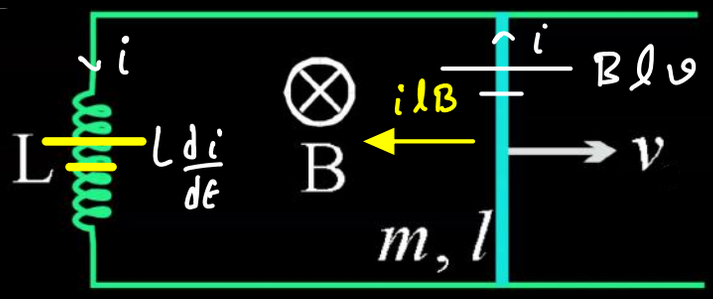
Consider an inductor slider arrangement shown in the figure. Suppose we impart a velocity v_0 to the slider at $t = 0$ and we wish to figure out its velocity and



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displacement as a function of time. We shall show that in absence of any resistance, the slider will execute a simple harmonic motion.

Now let velocity of slider be v at some instant. Then by loop law:



$$L\,Di - B\ell v = 0 \quad (1)$$

(D is differential operator d/dt)

Now applying LOM to the slider:

$$-i\ell B = m\,Dv \quad (2)$$

D(2) $\Rightarrow$

$$-B\ell\,Di = m\,D^2v \qquad \text{now } Di = B\ell v/L \text{ from (1)}$$

$$\Rightarrow -\frac{B^2\ell^2 v}{L} = m\frac{d^2v}{dt^2} \Rightarrow \frac{d^2v}{dt^2} = -\frac{B^2\ell^2}{mL}v$$

Thus velocity varies simple harmonically, also @ $t=0$, $v=v_0$ and $i=0 \Rightarrow dv/dt = 0$, thus velocity is starting from +ve extreme position of phasor thus:

$$v = v_0\cos\omega t \quad \text{where } \omega = B\ell/\sqrt{mL}$$

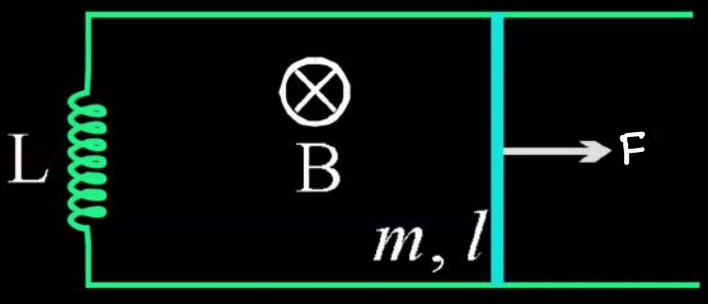
Now displacement: $s = \int_0^t v\,dt = \frac{v_0}{\omega}[\sin\omega t]_0^t = \frac{v_0}{\omega}\sin\omega t$ Ans.

Thus displacement is also simple harmonic.

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EXAMPLE:

In the inductor slider arrangement suppose a constant force F were applied with initial slider velocity being zero, find the slider velocity as a function of time.



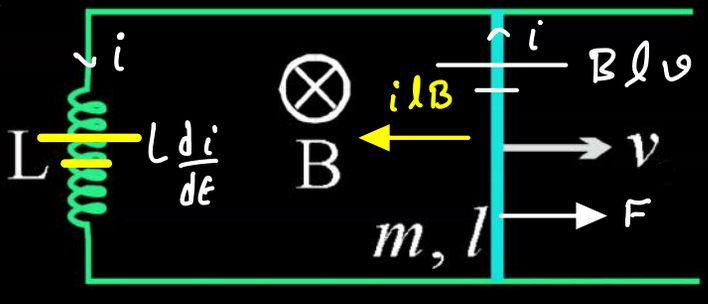
Solution: Loop law gives:

$$L\,Di = B\ell v \quad (1)$$

LOM gives:

$$F - i\ell B = m\,Dv \quad (2)$$

D(2) $\Rightarrow -B\ell\,Di = m\,D^2v$ (3)



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Substituting D_i from ①: $\quad -\frac{B^2\ell^2}{mL}v = \frac{d^2v}{dt^2}$

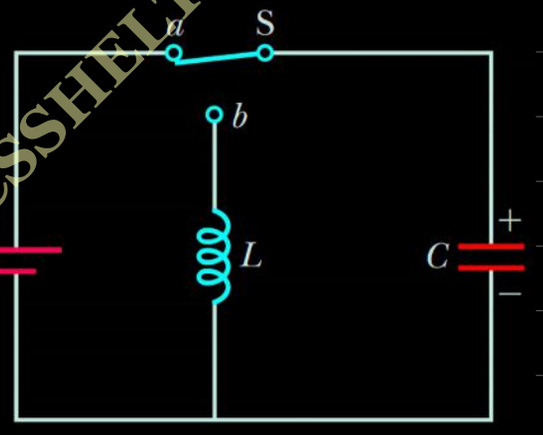
once again we have same differential equation. Now at $t=0$, $v=0$, $dv/dt = F/m$. Thus it is a case of phasor from mean position going rightward, thus, $v = v_0 \sin\omega t$ such that at $t=0$, $\frac{dv}{dt} = F/m$ i.e. $v_0\omega = F/m$ or $v_0 = \frac{F}{m\omega}$ thus $v = \frac{F}{m\omega}\sin(\omega t)$

where $\omega = B\ell/\sqrt{mL}$. Now $s = \int v\,dt = \frac{F}{m\omega^2}[-\cos\omega t]_0^t$

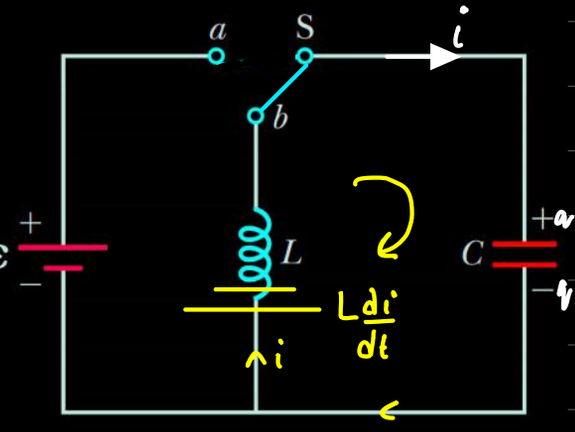
i.e $s = \frac{F}{m\omega^2}[1-\cos\omega t]$, thus, rod does SHM.

LC Oscillations:

Consider the circuit shown in the figure. Suppose once the capacitor is charged, we switch the connection to point b thus bringing the inductor in the loop. We shall show that this circuit is governed by SHM like differential equation and hence leads to oscillatory solutions.



The capacitor will begin to discharge through the inductor (we have shown the direction of current just so that we can write $i = dq/dt$). Now applying the loop law:



$$q/C + L\frac{di}{dt} = 0 \Rightarrow \frac{d^2q}{dt^2} = \frac{-1}{LC}q \quad ①$$

this is an SHM like differential equation with $\omega = 1/\sqrt{LC}$. Thus charge at any point of time can be simply written as $q = Q_{max}\cos(\omega t+\phi)$ where the value of Q_{max} will depend on initial values of charge (q) and current (dq/dt) in the capacitor.

Now current as a function of time, $i = dq/dt = -Q_{max}\omega\sin(\omega t+\phi)$

$\Rightarrow i = -I_{max}\sin(\omega t+\phi)$ where $I_{max} = Q_{max}\omega$.

Notice that equation ① looks just like $d^2x/dt^2 = -k/m\,x$

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whose solution is $x = A\cos(\omega t + \phi)$, thus by comparison charge looks like analog of position, also $\frac{1}{\sqrt{LC}} \longrightarrow \sqrt{\frac{k}{m}}$ thus mass looks like analog of inductance and Capacitance looks like analog of $1/k$. Now by extension $\frac{1}{2}Li^2$ must be analog of $\frac{1}{2}mv^2$ and $q^2/2C$ must be analog of $\frac{1}{2}kx^2$. In spring mass systems, we have $\frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \text{constant}$, let us see if a similar result holds for LC oscillatory circuit. Now:

$q = Q_{max}\cos(\omega t + \phi) \Rightarrow$ Electric Energy $U_E = q^2/2C = \frac{Q_{max}^2}{2C}\cos^2(\omega t + \phi)$

Now current $= -Q_{max}\omega \sin(\omega t + \phi)$ as described earlier thus,

$U_B = \frac{1}{2}Li^2 = \frac{1}{2}L Q_{max}^2 \omega^2 \sin^2(\omega t + \phi) = \frac{Q_{max}^2}{2C}\sin^2(\omega t + \phi)$

$(\because \omega^2 = 1/LC)$. Now $U_E + U_B = Q_{max}^2/2C$ which is a constant (this looks just like $\frac{1}{2}kA^2$).

→ Thus in LC oscillations sum of Electrical Energy and Magnetic Energy remains constant. Electric Energy is analog of potential Energy and magnetic energy is analog of kinetic Energy. Thus:

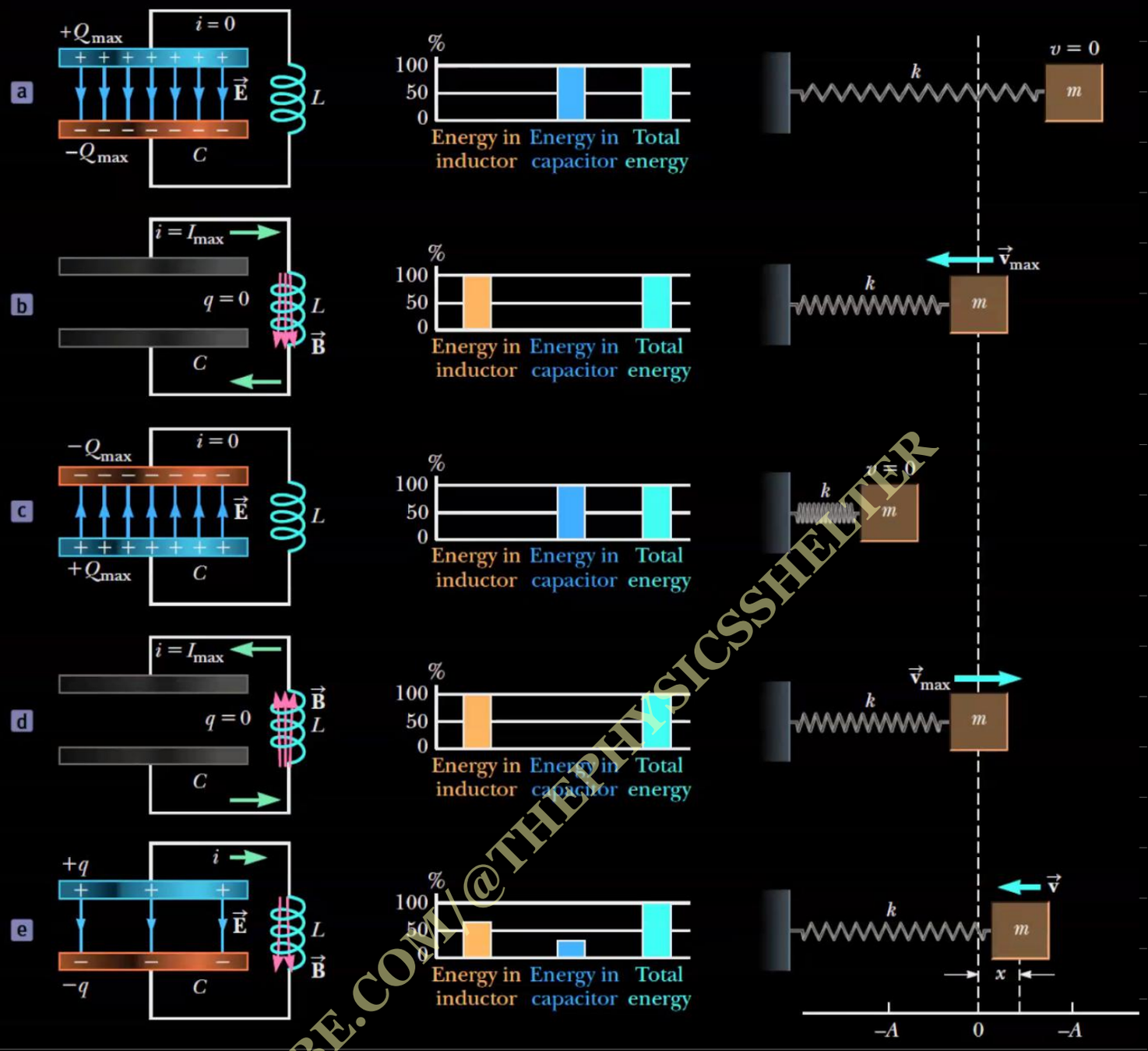
$$\boxed{\frac{Q_{max}^2}{2C} = \frac{1}{2}L I_{max}^2} \quad \left(\because \frac{1}{2}mv_{max}^2 = \frac{1}{2}kA^2\right).$$

→ Suppose initial current in the circuit is i and initial charge on the capacitor is q then maximum current and maximum capacitor charge can be readily found by energy conservation, thus:

$$\frac{1}{2}Li^2 + \frac{q^2}{2C} = \frac{Q_{max}^2}{2C} = \frac{1}{2}L I_{max}^2$$

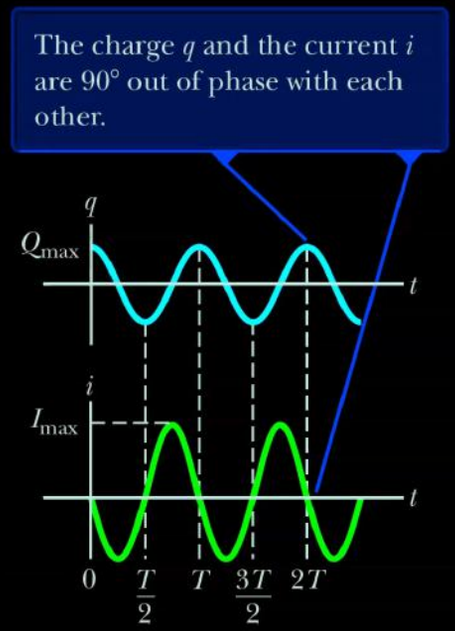
→ The spring mass analogy of LC circuit in terms of energy exchanges is shown in the next diagram. (Study it carefully):

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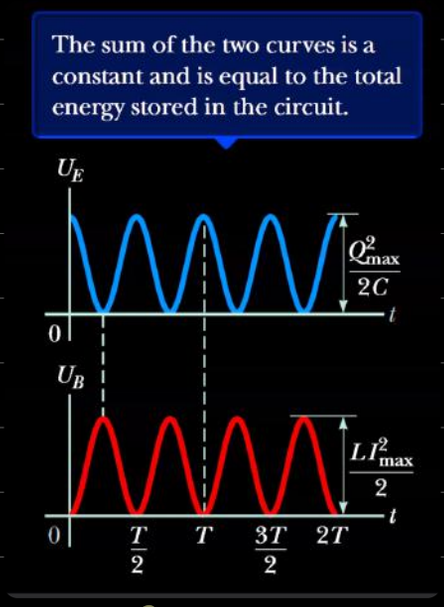


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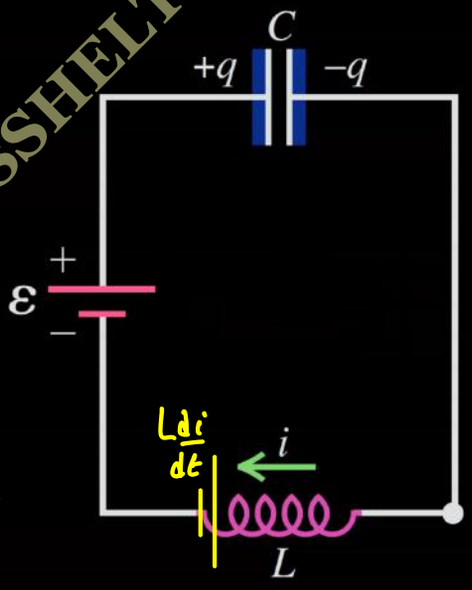
→ Current and charge are 90° out of phase.
Thus if $q = Q_{max} \cos \omega t$, $i = I_{max} \cos(\omega t + \pi/2)$
Thus to be precise, current leads ahead of the charge by a phase of $\pi/2$ (Remember we have considered current as dq/dt and not $-dq/dt$ otherwise current would appear to lag by a phase of $\pi/2$). The phase relation can also be seen graphically as represented in adjacent figure.



→ Electric Energy and Magnetic Energy are 180° out of phase with their sum being constant. Also these fluctuate with a time period of T/2 where T is the time period of oscillations of the charge and the current in the circuit. (Their sum always remains constant).
This can be graphically seen from the adjacent figure.



LC Oscillations With A Series Battery:



Consider the LC circuit shown in the adjacent figure. Applying loop law, we get:

$$q/C + L\frac{di}{dt} - \mathcal{E} = 0$$

$$\Rightarrow \quad \frac{d^2q}{dt^2} = \frac{1}{L}(\mathcal{E} - q/C) = -\frac{1}{LC}(q - \mathcal{E}C)$$

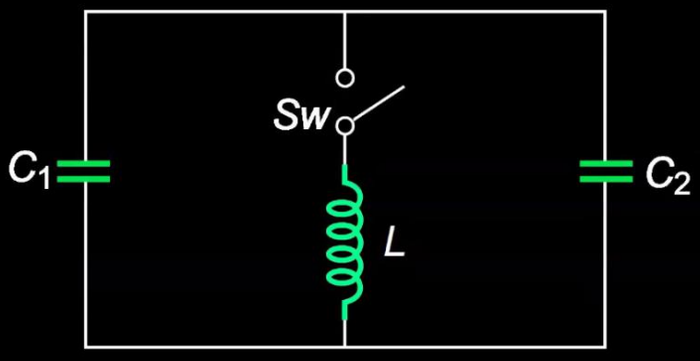
i.e $\frac{d^2q}{dt^2} = -\frac{1}{LC}(q - C\mathcal{E})$, now this equation is just like $\frac{d^2x}{dt^2} = -\omega^2(x - x_0)$ where x_0 represented the mean position wrt origin, similarly here $C\mathcal{E}$ represents the mean charge on the capacitor, with general solution for net charge on capacitor being $q = C\mathcal{E} + Q_{max}\cos(\omega t + \phi)$.

→ Battery is analog of a constant force ($x_0 = F/k$ similarly $q_0 = C\mathcal{E}$).

→ With multiple precharged capacitors in series, we can break them in to batteries and empty capacitors to find the additional charge and the superpose the additional charge on the initial charges to get the net charge (note that additional charge will be an oscillatory solution).

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Example: Irodov: 4.98



In an oscillating circuit shown in Figure, the coil inductance is equal to $L = 2.5\,\text{mH}$ and the capacitor have capacitances $C_1 = 2.0\,\mu\text{F}$ and $C_2 = 3.0\,\mu\text{F}$. The capacitors were charged to a voltage $V = 180\,\text{V}$, and then the switch *Sw* was closed. Find:

(a) the natural oscillation frequency;

(b) the peak value of the current flowing through the coil.

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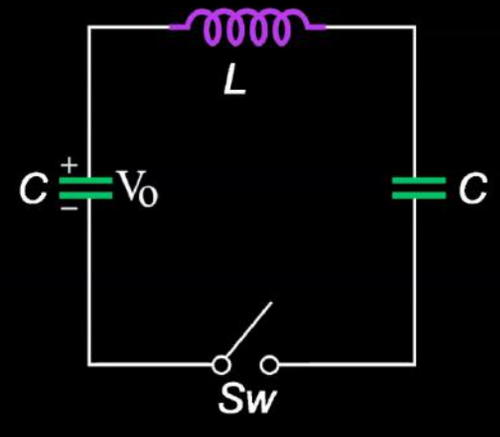
Solution: Capacitors are in || thus $C_{eff} = C_1 + C_2$, thus $\omega = \dfrac{1}{\sqrt{L(C_1+C_2)}}$ Ans.

Now for peak current, $\frac{1}{2} L I_{max}^2 = \frac{1}{2}(C_1+C_2)V^2$

$\Rightarrow I_{max} = V\sqrt{\dfrac{C_1+C_2}{L}}$ Ans. (Alternately $I_{max} = Q_{max}\omega$)

Numerical values: $T = 0.7\,\text{ms}$, $I_m = 8\,\text{A}$ Ans.

Example: Irodov 4.99:



An electric circuit shown in Figure, has a negligibly small active resistance. The left-hand capacitor was charged to a voltage V_0 and then at the moment $t = 0$ the switch *Sw* was closed. Find the time dependence of the voltages in left and right capacitors.

Solution: If we want we can do this problem by forming the differential equation. However we will here use the technique of breaking the precharged capacitor in to a battery and empty capacitor and comparing with a spring mass system. For our case: $C_{eff} = C/2$, $\mathcal{E} = V_0$ and initial charge is zero. This is just like a spring mass

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system released from natural length position under gravity. The solution to such a system is $x = \frac{mg}{k} - \frac{mg}{k}\cos(\omega t)$

(displacement amplitude is simply the difference of mean displacement and initial displacement where velocity is zero), thus by analogy, in our case, additional charge is given as:

$q_a = \frac{CV_0}{2}[1 - \cos\omega t]$, thus potential difference across right capacitor is simply $V_R = q_a/C = \frac{V_0}{2}(1 - \cos\omega t)$ Ans.

where $\omega = 1/\sqrt{L(C/2)} = \sqrt{2/LC}$. Similarly, for left capacitor. $V = V_0 - \frac{q_a}{C}$

$\Rightarrow V_L = V_0 - \frac{V_0}{2}(1 - \cos\omega t) = \frac{V_0}{2}[1 + \cos\omega t)$ Ans.



PERFECTLY CONDUCTING LOOP IN MAGNETIC FIELD: The net EMF in any perfectly conducting loop must be zero since any finite EMF would lead to infinite current. Thus flux through a perfectly conducting loop always remains constant. If an external magnetic field is changed or orientation of such a loop is changed in external field then a current is induced such that flux due to self inductance compensates for the loss of flux due to other changes.

EXAMPLE : IRODOV 4.100 :

An oscillating circuit consists of an inductance coil L and a capacitor with capacitance C. The resistance of the coil and the lead wires is negligible. The coil is placed in a permanent magnetic field so that the total flux passing through all the turns of the coil is equal to φ. At the moment $t = 0$ the magnetic field was switched off. Assuming the switching off time to be negligible compared to the natural oscillation period of the circuit, find the circuit current as a function of time t.

Solution: In the superconducting coil, flux can't change abruptly thus a current is induced to compensate of loss of flux due to switching off

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external magnetic field. Let this current be i_0 then: $Li_0 = \phi$
$\Rightarrow i_0 = \phi/L$. Thus we have an LC system with zero initial charge and initial current $i_0 = \phi/L$. This represents phasor mean position thus

$q = Q_{max} \sin(\omega t)$ where $Q_{max} = i_0/\omega$, thus,
$q = \frac{\phi}{L\omega} \sin(\omega t)$ and current is dq/dt i.e $\frac{\phi}{L} \cos\left(\frac{t}{\sqrt{LC}}\right)$ Ans.

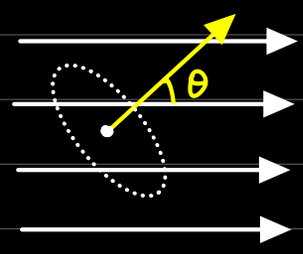
EXAMPLE: IRODOV 3.323:

A superconducting round ring of radius a and inductance L was located in a uniform magnetic field of induction B. The ring plane was perpendicular to the vector **B**, and the current in the ring was equal to zero. Then the ring was turned through 90° so that its plane became parallel to the field. Find:
(a) the current induced in the ring after the turn;
(b) the work performed during the turn.

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Solution: (a) By flux conservation, $L \times i = B \times \pi a^2 \Rightarrow i = \pi a^2 B/L$ Ans.

(b) Suppose the ring is turned through an angle θ then by flux conservation,
$B\pi a^2 = Li + B\cos\theta \pi a^2$ (1)
Now external torque to rotate the loop slowly must be exactly equal in magnitude to the magnetic torque.



Thus:
$\tau = i\pi a^2 B \sin\theta$, from (1), $i = \pi a^2 B(1-\cos\theta)/L$
Now work done by external agent:

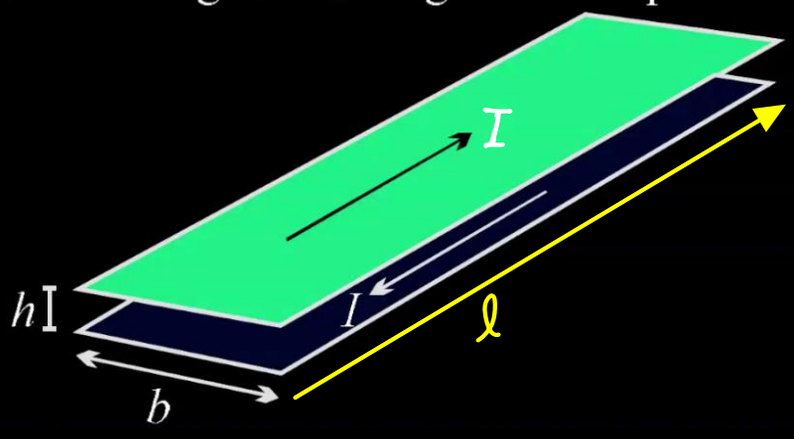
$$dW = \tau d\theta = \pi a^2 B \sin\theta \cdot \frac{\pi a^2 B}{L}(1-\cos\theta)\, d\theta$$

$$\Rightarrow W = \frac{\pi^2 a^4 B^2}{L} \int_0^{\pi/2} \sin\theta(1-\cos\theta)\, d\theta = \pi a^4 B^2/2L \text{ Ans.}$$

→ Note the failure of $-\vec{m}\cdot\vec{B}$ (the jugaad potential energy), here $\vec{m}\cdot\vec{B}$ is zero finally as well as initially, we could have used it only if $\vec{M}$ were constant throughout, here M was changing continuously).

EXAMPLE: IRODOV: 3.321:

Find inductance of a unit length of a long double tapeline as shown in figure.



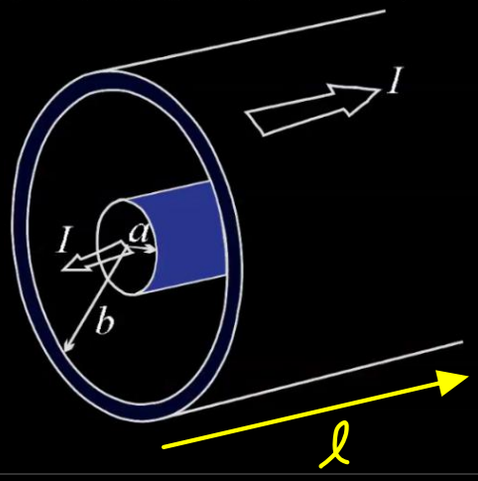
Solution: Whenever we have to find self inductance, we need to consider the direction of field lines. The flux is calculated by considering a cross section perpendicular to the field lines. For the double tape line, the field in the region between tapes is $\mu_0 \lambda$ (∵ field between capacitor plates is σ/ϵ_0). Thus $B = \mu_0 I/b$, These field lines are ⊥ to the rectangular cross section of area $h \times l$, thus,

$$\phi = \frac{\mu_0 I}{b} \times hl \Rightarrow L = \phi/I = \frac{\mu_0 hl}{b}, \text{ for } l = 1,$$

$L = \mu_0 h/b$ Ans.

EXAMPLE: IRODOV: 3.319:

A long coaxial cable consists of two thin walled cylinders with radii a & b. The current flows down the surface of inner cylinder and back along the outer cylinder. Find the self-inductance. Take $b/a = \eta$.



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Solution: Here field lines are circular centred on the axis of the system. The cross section perpendicular to the field lines is a rectangle of dimensions $(b-a) \times l$. Now magnetic field at a general distance from axis: $B = \mu_0 I / 2\pi r$. Now consider a strip of radial thickness dr from the axis. This has an area $A = l\,dr$ thus $d\phi = \frac{\mu_0 I}{2\pi r} \times l\,dr \Rightarrow \phi = \frac{\mu_0 I l}{2\pi} \int_a^b \frac{1}{r}\,dr$

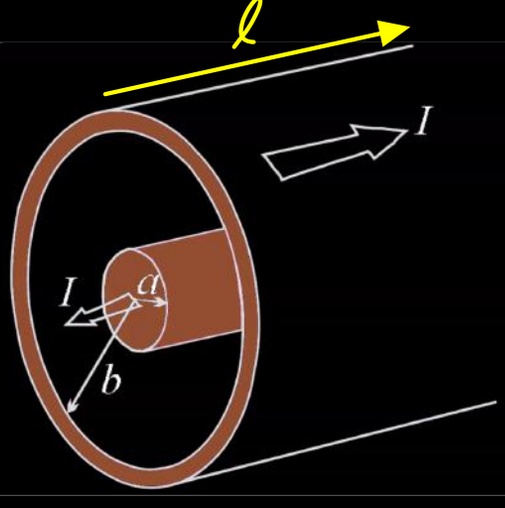
$\Rightarrow \phi = \frac{\mu_0 I l}{2\pi} \ln \eta \quad (\because b/a = \eta) \quad \Rightarrow L = \phi/I = \frac{\mu_0 l}{2\pi} \ln \eta$. Ans.

DIFFICULTY WITH FINDING INDUCTANCE USING ϕ/I:

So far in the examples we considered conductors which were literally thin and therefore there was practically no flux associated due to existence of magnetic field in the matrix of the conductors. Whenever there is magnetic field in the matrix of the conductors, defining a loop for calculating the flux becomes ambiguous (in $\phi = LI$, I was the current flowing along the loop). To find inductance in such cases we can't use $\phi = LI$, however we still have a hope because, magnetic energy is $U = \frac{1}{2} L i^2$ thus we can always find self inductance as $L = \sqrt{2U}/i$ where U is the magnetic energy stored in space. Recall that magnetic energy density in space is given as $u = B^2/2\mu_0$, thus U can be evaluated using the integral $U = \int u\,dV$.

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EXAMPLE: Find the self inductance of a coaxial cable. The inner conductor is a solid cylindrical rod having uniform current density per unit cross section, outer conductor is a thin cylindrical shell.



Solution: Here outside the system magnetic field is zero ($\because$ Ampere's law) and in the

Region inside the system, for $r < a$, $B = \mu_0 Jr/2$ and for region $a < r < b$, $B = \mu_0 I/2\pi r$.

Now let us find magnetic Energies region-wise. Consider an annular cylinder between radii r and $r+dr$:

For $r < a$,

$$U_I = \int dU = \int_0^a \frac{1}{2\mu_0} \times \left[\mu_0 \frac{(I/\pi a^2)\, r}{2}\right]^2 \times 2\pi r\, dr \times \ell$$

$$\Rightarrow U_I = \mu_0 I^2 \ell / 16\pi \quad , \text{ Now for outside region: } (a < r < b)$$

$$U_{II} = \int dU = \int \frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi r}\right)^2 \times 2\pi r\, dr\, \ell = \frac{\mu_0}{4\pi} I^2 \ell \ln(b/a)$$

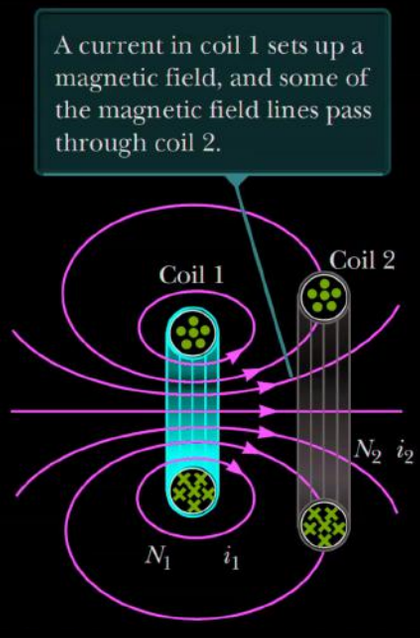
Thus $U = \frac{1}{2} L I^2 = \frac{\mu_0 I^2 \ell}{4\pi} \left[\frac{1}{4} + \ln(b/a)\right]$

$$\Rightarrow L = \frac{\mu_0 I^2 \ell}{4\pi} \left[\frac{1}{2} + 2\ln(b/a)\right] \text{ Ans.}$$

Mutual Inductance :

When two inductors or coils one in a close proximity to each other, the magnetic flux caused by current in one coil links with the other coil, thus inducing a voltage in the latter. This phenomenon is called mutual induction. If Φ_{21} is the flux linkage through coil 2 due to i_1 current in coil 1, then we define mutual inductance coefficient,

$M_{21} = \frac{\Phi_{21}}{i_1}$, Note that flux linkage is



flux per turn multiplied by number of turns. Similarly we can also define mutual inductance coefficient $M_{12} = \phi_{12}/i_2$

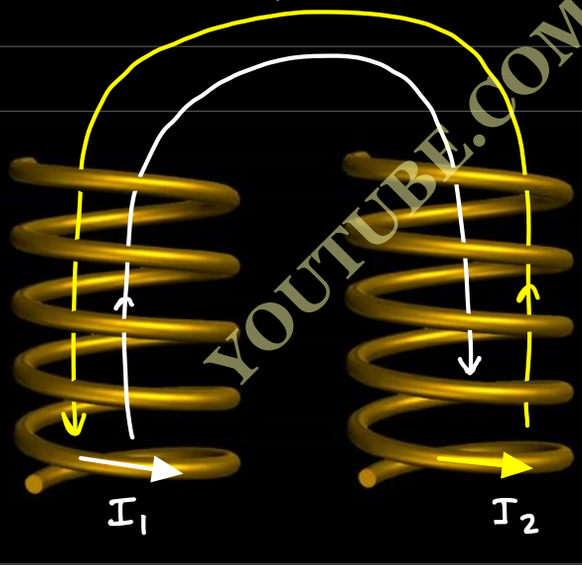
Later on we will show the reciprocity theorem of mutual inductance

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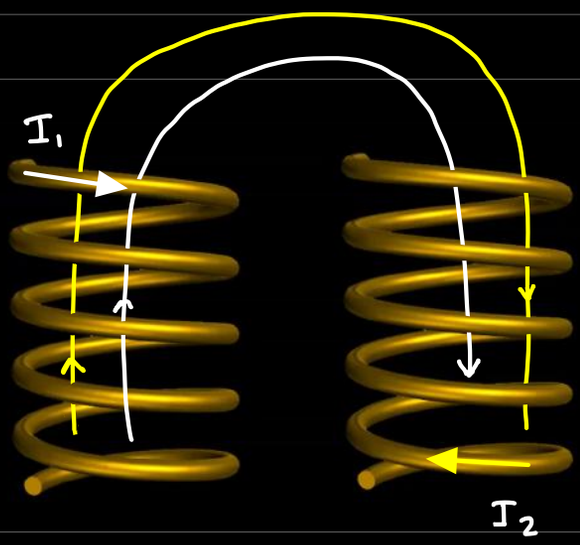
i.e $\boxed{M_{21} = \frac{\phi_{21}}{i_1} = M_{12} = \frac{\phi_{12}}{i_2}}$ therefore we can simply say $M_{21} = M_{12} = M$ (say).

→ The induced EMF in any coil due to the other is $\mathcal{E}_2 = M\frac{di_1}{dt}$ and $\mathcal{E}_1 = M\,di_2/dt$.

→ **Sign Of Mutual Inductance:** For deciding sign of mutual inductance between two coils first we will have to decide a positive sense of current in the two coils (just as we assume branch current directions in circuits, negative means it is opposite to assumed direction of current). Now if positive currents flow in the assumed directions, and flux due to self current and flux due to current in the other coil have same sign (or magnetic fields have same direction) then mutual inductance is positive. If the fields are in opposite directions, mutual inductance is negative. (Study the following examples for a better understanding):



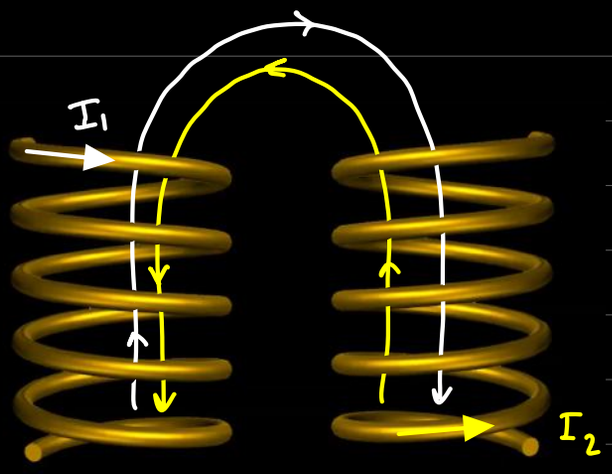
Self Field Opposite To Mutual Field
∴ M is Negative

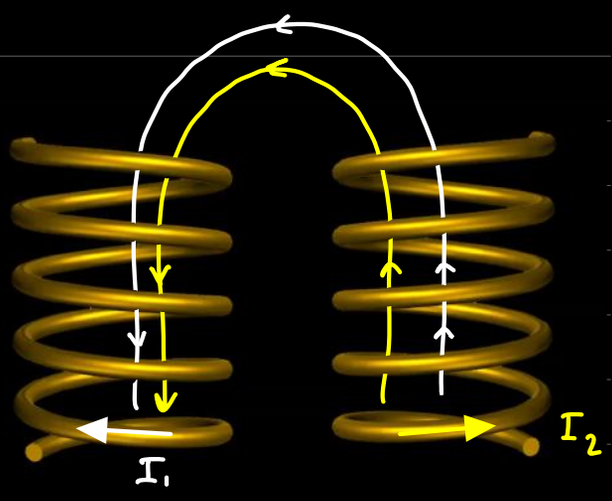


Self Field Along Mutual Field
∴ M Is Positive.

→ Note that coils in both figures are exactly same but assumption of current direction I_2 has changed thus changing the sign of M. Thus sign of M is a matter of convention and not an absolute fixed.

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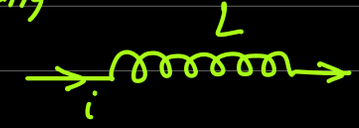




Opposite Directions Of Self & Mutual Fields ∴ M Is Negative

Same Direction Of Self & Mutual Field ∴ M Is Positive.

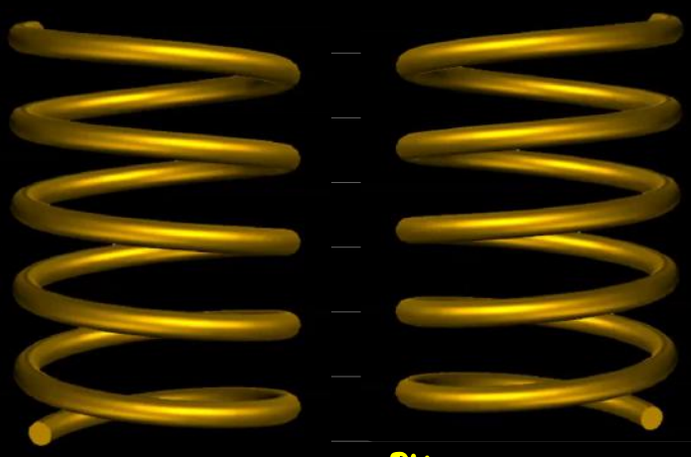
→ The method of sign convention (Inoldov's sign convention) requires one to know the 3D orientation of the coil and sense of rotation of current in the coil, in our examples we were able to comment on sign of mutual inductance because we had the 3D view of the Helices. However when drawing inductor as a circuit symbol, this will be ambiguous. For instance, in the followings figure, we can't be sure about direction of magnetic field lines. Hence L is only shown schematically as a circuit symbol. The field could be either rightward or leftward depending on sense of helix assumed. (Though generally right handed helices are more common, i.e as you advance from one end to the other end, turns follow the right hand rule i.e put thumb along the direction of traversal and curl of fingers gives the sense of Helix.).



→ If M is considered a signed value then total induced EMF in a coil can be written as $\mathcal{E}_1 = L\frac{dI_1}{dt} + M\frac{dI_2}{dt}$.

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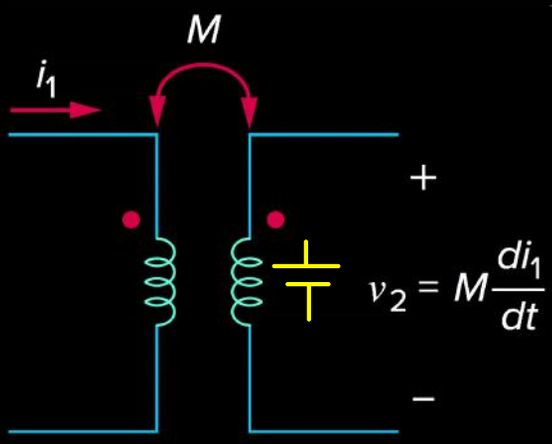
→ Study the adjacent figures to understand the difference between left handed and right handed helices.
A left handed helix always remains left handed whether you go from bottom to top or top to bottom.

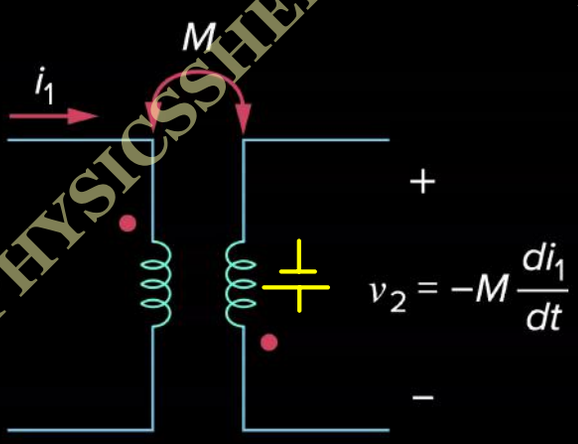


LEFT HANDED HELIX.

RIGHT HANDED HELIX

Dot Convention For Mutual Inductance :



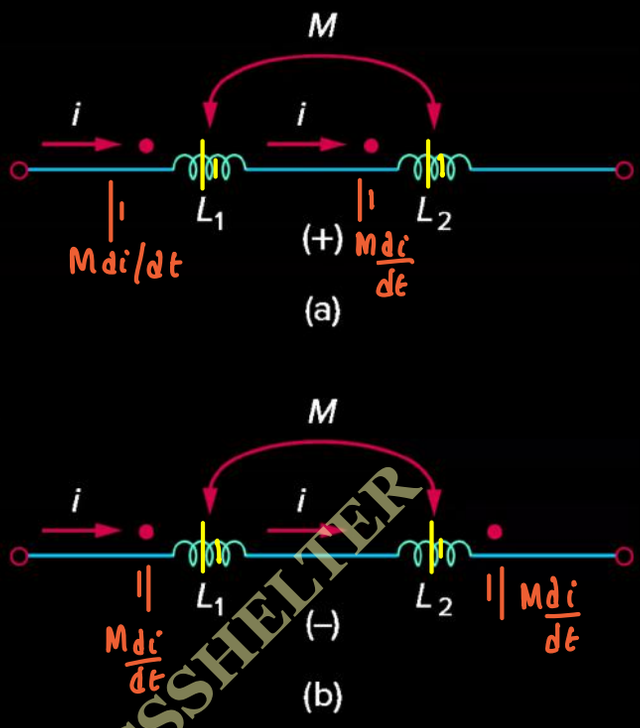


To avoid ambiguity about sign of mutual inductance in 2D circuit diagrams, electrical engineers put two dots on the coupled coils. When increasing current enters one dot, the other dot becomes the positive terminal of induced EMF battery. (Study the figures above). In this convention, M is always taken as positive and the ambiguity is removed due to presence of dots. (Note that if $\frac{di_1}{dt}$ were negative, it would automatically mean batteries in reversed polarities, however we need not worry about it while making differential equations for the circuit.

→ If the increasing current were entering from non-dotted side then in other coil, non-dotted side will be the positive terminal of induced EMF battery.

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Magnetically Coupled Inductors In Series:



As can be inferred from the adjacent figure, mutual inductance could be either opposing or supporting the self inductance, thus in case (a)

$$\Sigma = L\frac{di}{dt} + M\frac{di}{dt} + M\frac{di}{dt} + L\frac{di}{dt}$$

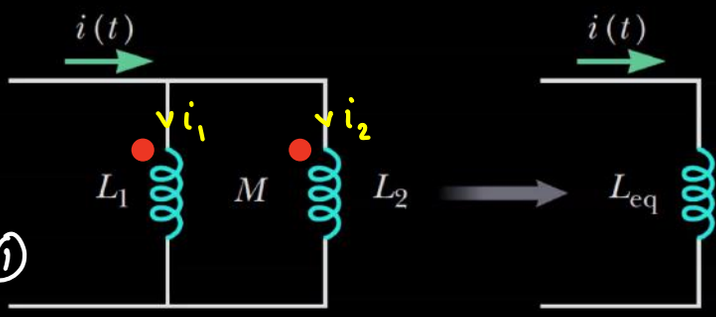
$$= L_{eff}\, di/dt \text{ thus,}$$

$$\boxed{L_{eff} = L_1 + L_2 + 2M} \leftarrow \text{(Series aiding)}$$

Similarly for series opposing:

$$\boxed{L_{eff} = L_1 + L_2 - 2M} \leftarrow \text{Series opposing (Engineering convention).}$$

Magnetically Coupled Inductors In Parallel:



Consider the circuit in adjacent figure.

$$\Sigma_{ind} = L_{eff}\,\dot{i} = L_1\dot{i}_1 + M\dot{i}_2 = L_2\dot{i}_2 + M\dot{i}_1 \quad (1)$$

Also $\dot{i}_1 + \dot{i}_2 = \dot{i}$ (2)

From equation (1), $(L_2 - M)\,\dot{i}_2 = (L_1 - M)\,\dot{i}_1$ (3)

From (2) & (3) $\dot{i}_1 = \dfrac{(L_2 - M)\,\dot{i}}{L_1 + L_2 - 2M}$, $\dot{i}_2 = \dfrac{(L_1 - M)\,\dot{i}}{L_1 + L_2 - 2M}$ (4)

Putting (4) in (1):

$$L_{eff}\,\dot{i} = L_1\left(\frac{L_2 - M}{L_1 + L_2 - 2M}\right)\dot{i} + M\left(\frac{L_1 - M}{L_1 + L_2 - 2M}\right)\dot{i}$$

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$$\Rightarrow \quad \boxed{L_{eff} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}} \leftarrow \text{Supporting Mutual Flux.}$$

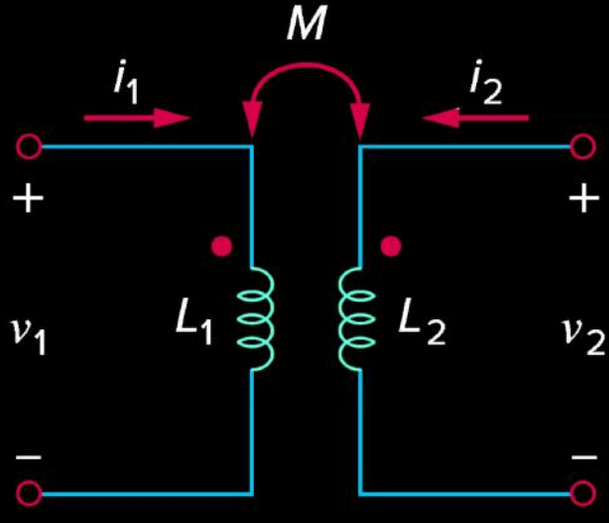
→ For inductors with opposing mutual flux we can simply replace M by $-M$. Thus:

$$\boxed{L_{eff} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}} \leftarrow \text{Opposing mutual flux.}$$

→ To satisfy Lenz law, Effective inductance must always be positive. Thus $L_1 L_2 - M^2 \geq 0 \Rightarrow M \leq \sqrt{L_1 L_2}$, thus mutual inductance of two coils can never be greater than geometric mean of self inductances. the ratio $k = M/\sqrt{L_1 L_2}$ is called coupling coefficient. $k \leq 1$
When every field line associated with first coil also passes through second coil and vice versa, then coupling coefficient is exactly 1.

Energy In A Coupled Circuit And Proof Of Reciprocity:

Considered coupled inductors shown in the figure. We initially assume i_1 and i_2 to be zero so that initial magnetic energy in space is zero. Now suppose we build current i_1 in coil 1 while maintaining zero current in coil 2 then power in coil 1: $p_1 = v_1 i_1 = L_1 \frac{di_1}{dt} i_1$



$\Rightarrow$ Total Energy stored for building $i_1 = \int p_1 dt$
$\Rightarrow U_1 = \frac{1}{2} L_1 i_1^2$. Now keeping i_1 as constant, we build a current i_2 in coil 2, however in coil 1, there will be a mutual induced EMF $M_{12} \frac{di_2}{dt}$ and a battery must work against it to maintain i_1 constant, also a battery in coil 2 is required for buiding up i_2. Combined power of coils in this step:

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$$P_2 = M_{12}\frac{di_2}{dt} \times i_1 + L_2\frac{di_2}{dt} \times i_2$$

$\Rightarrow$ Energy stored in this step, $U_2 = \int P_2\, dt = M_{12} i_1 i_2 + \frac{1}{2} L_2 i_2^2$

Now total energy stored in entire process: $U = U_1 + U_2$ thus:

$U = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M_{12} i_1 i_2$ ①, now we could also achieve the same final configuration by changing the order of operations, i.e first building i_2 and then building i_1 keeping i_2 constant, then:

$U = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M_{21} i_1 i_2$ ②

But since finally achieved configuration is same thus from ① & ②:

$M_{12} = M_{21} = M$ QED.

Also $\boxed{U = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M i_1 i_2}$ ← M is a signed value here.

→ This expression can be thought of as sum of self energies and interaction energies of the coils with $M i_1 i_2$ term representing the interaction energy.

→ If $\vec{B}_1$ is field due to coil 1 and $\vec{B}_2$ is field due to coil 2, it can be shown that interaction energy of coils is simply $\frac{1}{\mu_0}\int \vec{B}_1 \cdot \vec{B}_2\, dV$

This can be shown as follows:

$$U = \frac{1}{2\mu_0}\int (\vec{B}_1 + \vec{B}_2)^2\, dV = \frac{1}{2\mu_0}\int (\vec{B}_1 + \vec{B}_2)\cdot(\vec{B}_1 + \vec{B}_2)\, dV$$

$$= \int \frac{B_1^2}{2\mu_0} dV + \int \frac{B_2^2}{2\mu_0} dV + \frac{1}{\mu_0}\int \vec{B}_1 \cdot \vec{B}_2\, dV$$

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clearly the first and second terms represent self energies of the two coils, therefore the third terms must be interaction Energy.

→ From above discussion it is evident that in case of three coils, magnetic energy will be given as:

$$U = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + \frac{1}{2} L_3 i_3^2 + M_{12} i_1 i_2 + M_{13} i_1 i_3 + M_{23} i_2 i_3$$

[(can be extended to more inductors as well).

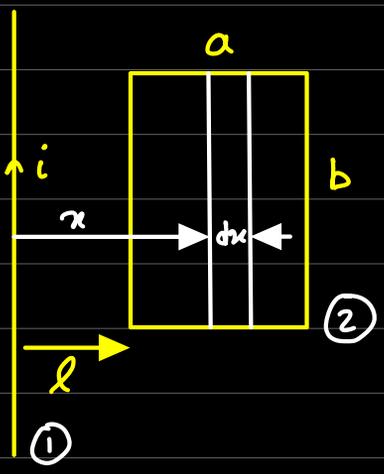
EXAMPLE: IRODOV: 3.329:

Calculate the mutual inductance of a long straight wire and a rectangular frame with sides *a* and *b*. The frame and the wire lie in the same plane, with the side *b* being closest to the wire, separated by a distance *l* from it and oriented parallel to it.

Solution: Consider the adjacent figure.

$$d\phi_{21} = \frac{\mu_0 i}{2\pi x} b\,dx \Rightarrow \phi_{21} = \frac{\mu_0 i b}{2\pi} \ln\left(\frac{l+a}{l}\right)$$

Now $M = \phi_{21}/i = \frac{\mu_0 b}{2\pi} \ln\left(\frac{l+a}{l}\right)$ Ans.



Think: Suppose in rectangular loop a current *i* were flown, what would be the flux through semi-infinite plane to the left of the infinite wire?

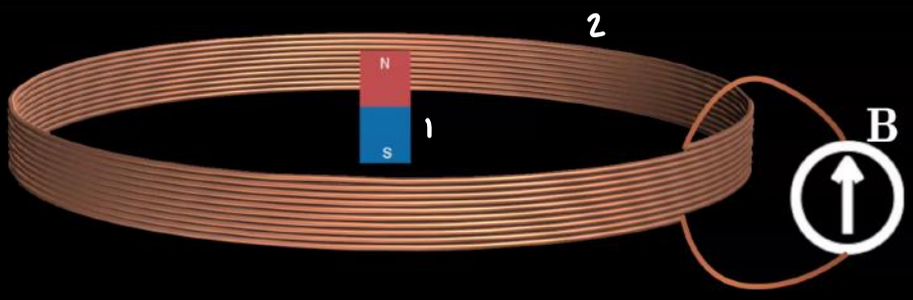
Solution: Imagine completing the boundary of semi-infinite plane. The mutual inductance of such a loop with the rectangular loop is same as the one we just calculated. Now $\phi_{12}/i_2 = \phi_{21}/i_1 \Rightarrow \phi_{12} = \frac{\phi_{21} \times i_2}{i_1} = M i_2$

Thus required flux: $\phi_{12} = \frac{\mu_0 b}{2\pi} \ln\left(\frac{l+a}{l}\right) i$ Ans.

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Example: Irodov 3.332:

A small cylindrical magnet, (See figure) is placed in the centre of a thin coil of radius a consisting of N turns. The coil is connected to a ballistic galvanometer B. The active resistance of the whole circuit is equal to R. Find the magnetic moment of the magnet if its removal from the coil results in a charge q flowing through the galvanometer. (Think of magnet as a small current carrying loop (or point magnetic dipole))



Solution: Let us imagine the magnet as a small magnetic dipole of radius $r << a$. Now mutual inductance:

$$M = \left(\frac{\mu_0 N i}{2a} \times \pi r^2\right) \Big/ i = \frac{\mu_0 N \pi r^2}{2a} \quad (\because \phi_{12}/i_2)$$

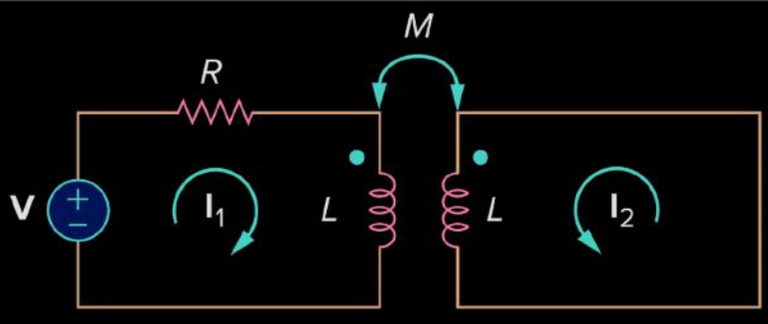
Now let magnetic moment of the magnet be $m = I\pi r^2$, then flux through the coil due to magnet, $\phi = M \times I$

$$\Rightarrow \phi = \frac{\mu_0 N \pi r^2}{2a} \times I = \frac{\mu_0 N m}{2a}, \text{ Now charge passed } q = \phi/R$$

Thus $$q = \frac{\mu_0 N m}{2aR} \Rightarrow m = \frac{2aRq}{\mu_0 N} \quad \text{Ans.}$$

Example:

For the adjacent circuit, develop a first order differential equation governing the current I_1 in the left loop.



Solution: Loop law for left loop:

$$I_1 R + L DI_1 + M DI_2 - V = 0 \quad (1)$$

$$M DI_1 + L DI_2 = 0 \quad (2) \longleftarrow \text{for right loop.}$$

$(D: d/dt)$

Now $L \times (1) - M \times (2)$:

$$I_1 LR + (L^2 - M^2) D I_1 - VL = 0 \Rightarrow (L^2 - M^2)\frac{dI_1}{dt} = L(V - I_1 R)$$

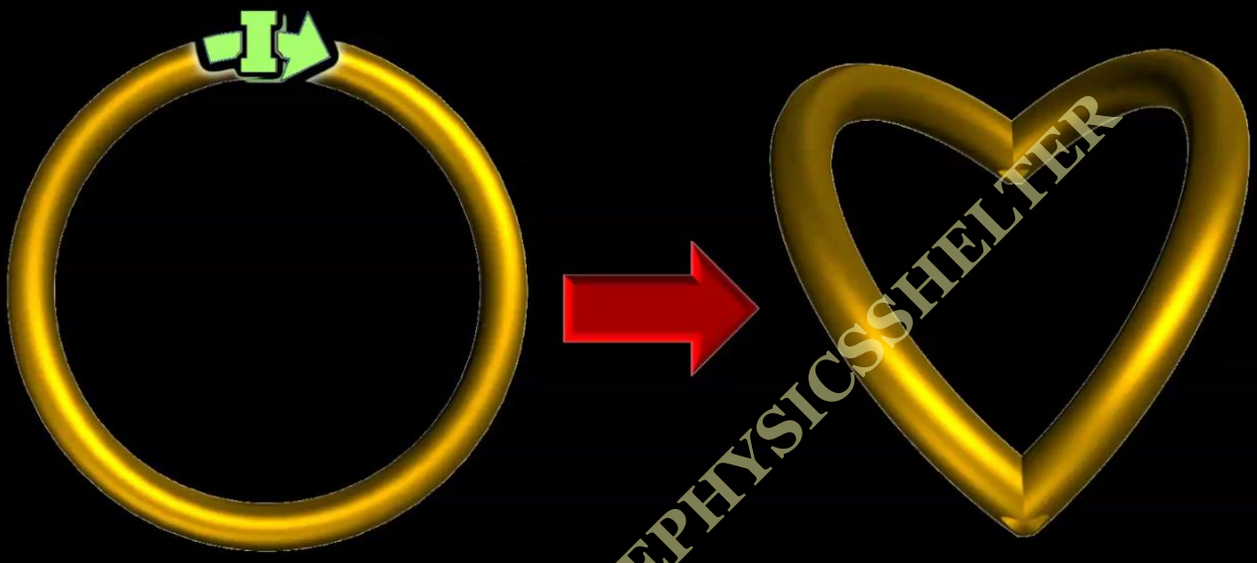
Ans.

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EXAMPLE :

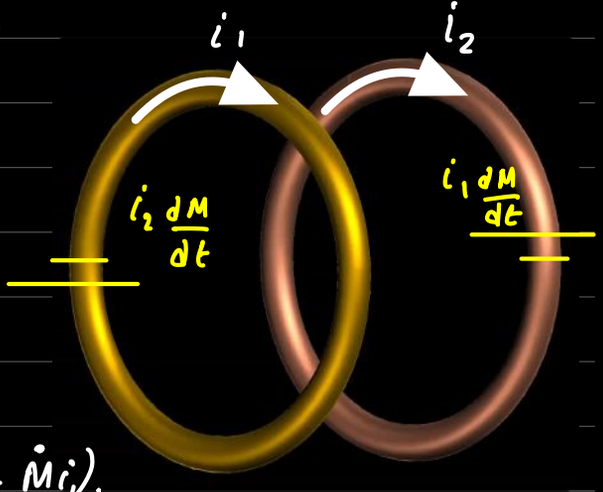
ORIGINAL INDUCTOR BENDING CHALLENGE BY AMBARISH

You bent a perfectly conducting thin circular current carrying coil through an angle of 90° while a constant current was maintained through it. The change in magnetic potential energy stored in the entire space in the process was-2 Joules. What was the work done by you in bending the coil. (Assume that work is done only against electromagnetic forces, ignore the effects of material rigidity/ elastic moduli/ gravity etc.).



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Solution: Before we do this problem let us look at a generalized situation. Consider two coils maintaining constant currents while changing relative orientation. Let $\dot{M}$ be the rate of change of mutual inductance, now due to changing mutual inductance, an EMF is induced in both coils even though currents are constant. ($\because \phi_{21} = M i_1$ thus $d\phi_{21}/dt = (\dot{M} i_1 + i_1 \dot{M})$, $i_1 = 0 \Rightarrow d\phi_{21}/dt = \dot{M} i_1$).



Now to maintain constant currents in the rings, external batteries must exactly produce an EMF to cancel the EMFs produced due to changing mutual inductances. Let ξ_1 and ξ_2 be the required external EMFs for this purpose. Then power developed by external EMF on loop 1 : $P_1 = \xi_1 i_1 = i_2 \frac{dM}{dt} \times i_1$

Thus if mutual inductance changed by an amount ΔM, the work done by this battery, $\Delta W_1 = \int p_1 \, dt = i_1 i_2 \Delta M$

$\Rightarrow$

$$\boxed{\Delta W_1 = i_1 i_2 \Delta M}$$

Similarly work done by battery in other loop is also $\Delta W_2 = i_1 i_2 \Delta M$.

Now:

$$U = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M i_1 i_2$$

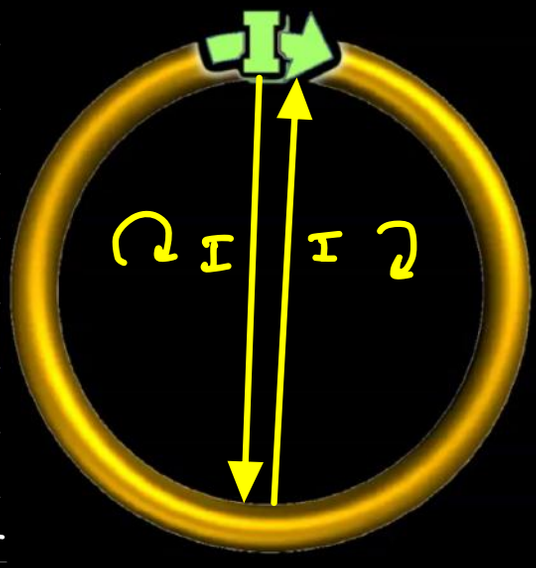
$\Rightarrow$ $\Delta U = i_1 i_2 \Delta M = \Delta W_1 + \Delta W_2 + \Delta W_{ext} = 2 i_1 i_2 \Delta M + \Delta W_{ext}$

$\Rightarrow$ $\Delta W_{ext} = -i_1 i_2 \Delta M$. Thus if constant currents are maintained, we have the relation:

$$\boxed{\Delta U = \Delta W_1 = \Delta W_2 = -\Delta W_{ext}}$$

Also, $\dot{U} = p_1 = p_2 = -p_{ext}$.

Now coming to current problem:
We can consider the loop to be made of two semicircular coils and bending the loop changes the mutual inductance of the two coils. Now change in magnetic potential energy is given as:
$\Delta U = -2$ Joules, thus according to our analysis, $\Delta W_{ext} = -\Delta U$ thus, work done by external agent, $\Delta W_{ext} = +2$ J Ans.



→ Think: what is the work done by the battery to maintain the current?
Solution: Here both batteries are in same loop itself thus $\Delta W_{bat} = \Delta W_1 + \Delta W_2$
$= \Delta U \times 2 = -4$ Joules Ans.

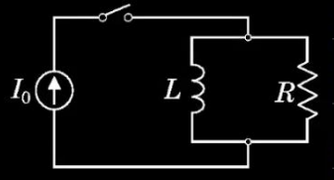
→ In our argument, exact shape of the coil does not matter, nor does exact nature of bending matter. Our answer would have

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been the same even if coil were of some other shape and bent arbitrarily. (Or for that matter there were two coils whose relative orientation were changed while maintaining their currents).

EXAMPLE: PATHFINDER: MC09:

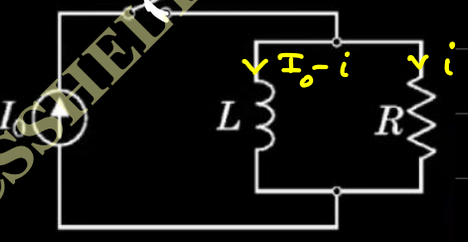
A circuit shown consists of a coil of inductance L, an unknown resistance R, a switch and an ideal current source of current I_0. Find total heat dissipated in the resistance after the switch is closed.



(a) Zero (b) $\frac{1}{4}LI_0^2$

(c) $\frac{1}{2}LI_0^2$ (d) LI_0^2

Solution: Refer to the adjacent figure:



$$L\frac{d}{dt}(I_0 - i) = iR$$

$$\Rightarrow -L\frac{di}{dt} = iR \text{, Now multiply } i \text{ on}$$

both sides: $-Li\frac{di}{dt} = i^2R \Rightarrow \int_0^\infty i^2R\,dt = \int_{I_0}^{0} -Li\,di = \frac{1}{2}LI_0^2$ Ans. (C)

(Note the limits of i, i is the current through the resistor, initial all current passes through resistor (∵ inductor acts like open circuit). Finally all current passes through inductor).

METHOD 2: Thevenizing the inductor $\mathcal{E}_{th} = I_0R$, $R_{th} = R \Rightarrow \tau = L/R$

Now current in any branch, $i = i_\infty + (i_0 - i_\infty)e^{-t/\tau}$, thus current through resistor: $i = I_0e^{-tR/L}$. Now heat dissipated:

$$H = \int_0^\infty i^2R\,dt = \frac{1}{2}LI_0^2 \text{ Ans.}$$

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PATHFINDER MCQ 11:

Consider a series combination of a resistance R, an inductance L, and a capacitance C undergoing electromagnetic oscillations. During a certain period, amplitude of current through the resistance drops from I to $0.5I$. What is the amount of heat dissipated by the resistance in this period?

(a) $0.375LI^2$ (b) $0.375I^2CR^2$

(c) $0.375I^2(L+CR^2)$ (d) $0.375LI^2+0.75I^2CR^2$

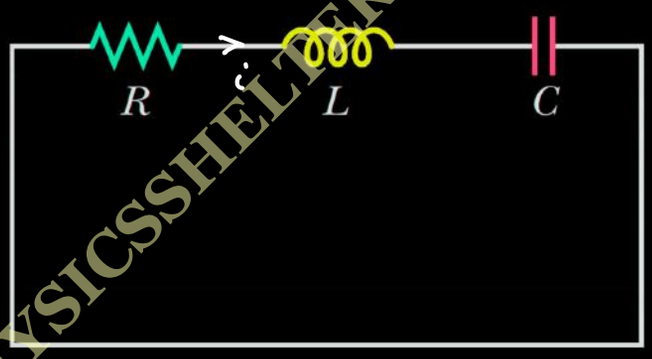
(Consider I and $0.5I$ currents to be maximum current points).

Solution: Differential equation governing LCR circuit is:

$$iR + L\,di/dt + q/C = 0 \quad (1)$$

or

$$L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{1}{C}i = 0$$



(Very similar to: $m\ddot{x} + b\dot{x} + kx = 0$), thus its solution is of the form, $i = I_0 e^{-\beta t}[\cos\omega t + \phi]$. Note that due to exponential decaying term, current maxima do not occur when $\cos(\omega t+\phi) = \pm 1$, rather it will occur when $dI/dt = 0$, product rule will yield weird looking equation for locating peaks of the current). Now when current hit a peak value, $di/dt = 0$ thus from equation (1), $q = -i/Rc$.

Now when peak current is I, $q = -I/Rc$ and when it is $0.5I$, $q = -0.5I/Rc$, Thus heat lost = $U_i - U_f$ (loss in total PE)

thus:

$$\Delta H = \left[\tfrac{1}{2}LI^2 + \tfrac{1}{2}C(-I/RC)^2\right] - \left[\tfrac{1}{2}L(0.5I)^2 + \tfrac{1}{2}C(-0.5I/RC)^2\right]$$

$$= 0.375(L+CR^2)I^2 \quad \text{i.e option C Ans.}$$

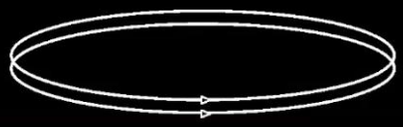
→ Recall damped oscillation equation: $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$. It has a solution of the form:

$$x = Ae^{-\beta t}\cos(\omega t+\phi) \text{ where } \omega = \sqrt{\omega_0^2 - \beta^2}$$

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EXAMPLE: PATHFINDER: BUILD 24:

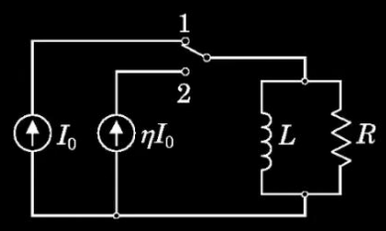
Two identical perfectly conducting rings are coaxially placed very close (infinitesimally small separation) to each other. Each of the coils has of inductance L and carries a current I_0 in the same direction. How much minimum amount of work has to be done in making their separation infinitely large?



Solution: Initially, the stored energy in space is $\frac{1}{2}LI_0^2 + \frac{1}{2}LI_0^2 + MI_0^2$ Here ∵ rings are very close together ∴ $M = L$ thus: $U_i = 2LI_0^2$. Now superconducting rings do not allow change in flux, initial flux through each ring, $\phi_i = LI_0 + MI_0 = 2LI_0$ thus at infinite separation if current is i then, $Li = 2LI_0$ $\Rightarrow i = 2I_0$. Thus $U_f = \frac{1}{2}L(2I_0)^2 \times 2 = 4LI_0^2$. Thus work done by external agent, $\Delta W = \Delta U = 4LI_0^2 - 2LI_0^2 = 2LI_0^2$ Ans.

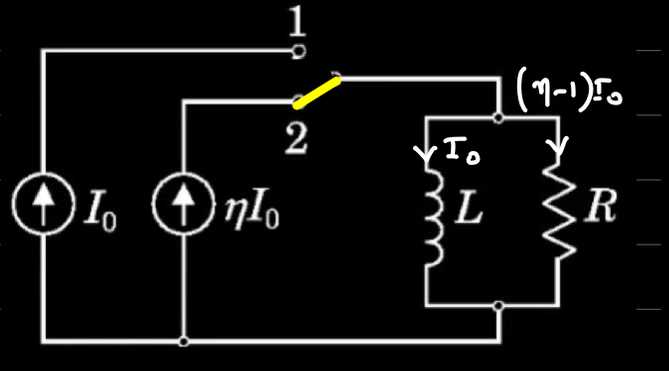
PATHFINDER: BUILD 29:

A parallel combination of an inductance L and a resistance R can be connected to either of the two ideal current sources of current I_0 and ηI_0 through a switch as shown. Initially the switch is in position 1 for a long time. Now the switch is thrown to position 2. Find amount of charge that will pass through the resistance and heat dissipated in it after the switching until a steady state is established.



Solution: Recall that at $t = \infty$ inductor behaves like short circuit and it does not allow sudden changes in current. Thus immediately after switching the currents are as shown in the figure. At $t = \infty$, entire ηI_0 will pass through the inductor. Now thevenizing the inductor, $\mathcal{E}_{th} = \eta I_0 R$ and $R_{th} = R$, thus time constant of the circuit is L/R. Thus current through resistor at any general time: $i = i_\infty + (i_0 - i_\infty)e^{-t/\tau}$ $\Rightarrow i = 0 + (\eta - 1)I_0 e^{-tR/L}$, Thus total charge passed $= \int i\,dt$

$= (\eta - 1)I_0 \int_0^\infty e^{-tR/L} = (\eta - 1)I_0 L/R$ Ans.

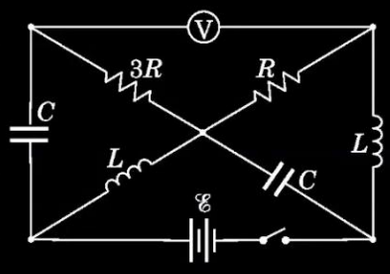


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Now total heat dissipated, $H = \int i^2 R\,dt = (\eta-1)^2 I_0^2 R \int_0^\infty e^{-2tR/L}\,dt$

$\Rightarrow H = \frac{1}{2} L (\eta-1)^2 I_0^2$ Ans.

EXAMPLE: PATHFINDER: BUILD 33:



A circuit shown consists of two resistances R and $3R$, two capacitors each of capacitance C, two inductors each of inductance L, an ideal battery of electromotive force $\mathscr{E}$, an ideal voltmeter and a switch. Initially the switch is closed and the circuit is in steady state.

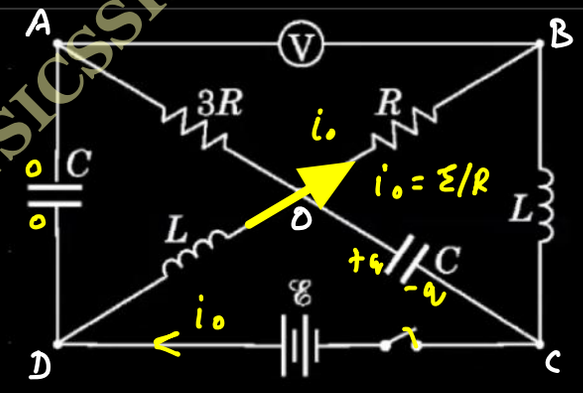
(a) Find the voltmeter reading.

Now, the switch is opened.

(b) Find the voltmeter reading immediately after the switching.

(c) Find the total heat dissipated in the resistors after the switching.

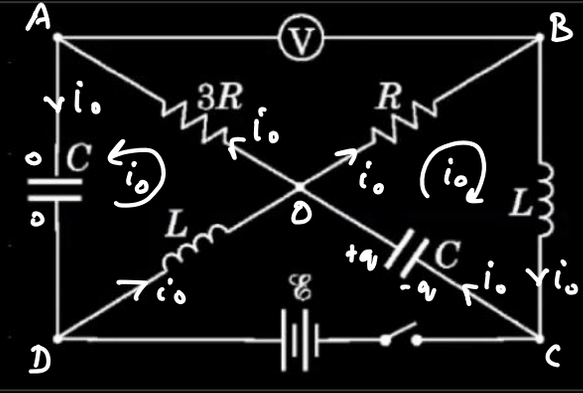
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Solution: In steady state, inductors behave like short circuits and capacitors like open circuits, thus before opening the switch the currents and charges on capacitors are as shown in the figure. The charge on capacitor on branch OC, $q = i_0 R \times C = C\mathscr{E}$, O and A are equipotential thus potential drop along AB is same as that along OB i.e $i_0 R = \mathscr{E}$ thus initial voltmeter reading before opening the switch is $\mathscr{E}$ Ans (a) (A is at higher potential than B).

Now immediately after opening the switch, inductors do not "want" to change their currents suddenly, accordingly currents are flowing as shown in adjacent figure. Thus total potential drop from B to A can be calculated along path BOA as: $V_B - V_A = -i_0 R + i_0 \times 3R = 2i_0 R = 2\mathscr{E}$

Thus $V_B - V_A = 2\mathscr{E}$ Ans (b).



Now after opening the switch LCR oscillations will continue in loops OBC and DOA until entire stored energy is dissipated.

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(or possibly overdamped solution, if R is sufficiently large).

Thus heat dissipated : $\frac{1}{2} L i_0^2 \times 2 + q^2/2C = \frac{1}{2} L \left(\frac{\mathcal{E}}{R}\right)^2 \times 2 + \frac{1}{2} C \mathcal{E}^2$

$\Rightarrow H = L\mathcal{E}^2/R + \frac{1}{2} C \mathcal{E}^2$ Ans (c).

We can also find the heats dissipated in individual resistors by finding energy stored in the two loops separately. Thus:

$H_{3R} = \frac{1}{2} L i_0^2 = \frac{1}{2} L \mathcal{E}^2/R^2$ and $H_R = \frac{1}{2} L \left(\frac{\mathcal{E}}{R}\right)^2 + \frac{1}{2} C \mathcal{E}^2$ Ans.

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